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Research areas	Trustworthy AI and foundation-model interpretability; reasoning and inference-time compute; agentic AI; multimodal foundation models; AI for science
Publications	111 peer-reviewed papers (58 journal, 53 conference), 83 under review, 3 book reviews
Citations	951 citations, *h*-index 17, *i*10-index 31 (*Google Scholar, November 14, 2026*)
Mentoring	16 postdoctoral and graduate researchers, 6 undergraduate researchers, 38 capstone students
Funding	6 external competitive awards as Lead PI or PI, 2 national programs (PI), 9 institutional awards

Biography. Hasan Kurban is an Assistant Professor in the College of Science and Engineering at Hamad Bin Khalifa University and Founding Director of the *Kurban Intelligence Lab*. His research develops statistical foundations for evaluating machine learning systems whose outputs can no longer be checked by hand. The organizing question is identifiability: which quantities an evaluation or learning protocol recovers, and which it only appears to. Where possible, each negative result is paired with a repair: a bias-corrected estimator, a blind baseline, or a distribution-free certificate.

In LLM evaluation, his group has shown that agent-as-a-judge verdicts shift with prompt language enough to invert the ranking of judge models (*COLM* 2026), and SAFE (*EMNLP* 2025, Findings) uses sparse autoencoders to mitigate hallucination.

The same standard extends to perception and science. IRIS (*ECCV* 2026), a real-world video benchmark of dynamical systems with known governing equations, exposed a gradient-flow bug in the state-of-the-art identification pipeline, and GRAPE (*NeurIPS* 2026) adds a concept-mismatch safety check to medical image diagnosis. In materials, CliffordIP (*npj Computational Materials*) builds $\mathrm{O}(3)$-equivariant interatomic potentials from Clifford algebra, and a *KDD* 2026 study shows that, for well-behaved models, out-of-distribution error under nanoparticle size extrapolation follows a power law forecastable from the in-distribution fit.

He serves on the editorial board of *Scientific Reports*, reviews for top AI venues including *NeurIPS*, *ICML*, *ICLR*, and *AAAI*, and organizes a workshop at The Web Conference and a shared task at *EMNLP*. Students are first authors on the group's main-program papers; alumni hold positions at Meta, ServiceNow, Eli Lilly, and the University of Maryland.

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RESEARCH INTERESTS

Each area is stated as the question that drives it.

- **Trustworthy AI and Foundation Model Interpretability.** *What does an evaluation protocol actually identify, and when can a model's output be certified?* Identifiability and corrected estimators for paired and multilingual LLM evaluation; hallucination detection and mitigation; sparse-autoencoder feature analysis and model editing; calibration, selective prediction, and abstention; jailbreak robustness and red-teaming.
- **Reasoning and Inference-Time Compute.** *When does additional test-time computation buy accuracy rather than the appearance of it?* Reinforcement learning with verifiable rewards (RLVR); posterior-constrained inference for mathematical reasoning; decoding-format confounds in consistency-based selection; multi-agent and neurosymbolic reasoning; robustness of chain-of-thought under noisy, conflicting, and adversarial evidence.
- **Agentic AI and LLM Agents.** *How do errors compound over long horizons, and what does a sound audit cost?* Tool use and long-horizon planning; retrieval-augmented and knowledge-grounded reasoning; agent-as-a-judge evaluation and its language and backbone sensitivity; cost-aware audit protocols; agent safety; scientific and biomedical discovery agents.
- **Multimodal Foundation Models.** *Do multimodal systems recover the structure of the world, or only correlations in the training distribution?* Vision–language models; video understanding and long-context temporal reasoning; world models and inverse identification of physical dynamics from video; conformal coverage for temporal grounding; embodied and spatial reasoning.
- **AI for Science.** *Can a learned model make a scientific claim that survives physical verification?* Equivariant and symmetry-constrained interatomic potentials; physics-informed learning; statistically valid screening under a limited first-principles budget; multimodal benchmarks and active learning for materials informatics, quantum chemistry, biomedical imaging, and energy systems.

SELECTED PUBLICATIONS

17 representative works, drawn from the 111 peer-reviewed papers published or accepted, which are listed in full below. **Bold** denotes Dr. Kurban, a dagger (†) the corresponding author, an asterisk (*) equal contribution, and underlining a student or postdoctoral researcher supervised or co-supervised by Dr. Kurban. Workshop and special-track papers are labeled with their track.

AI Conferences, Main Technical Program

S1 **GRAPE: Graph-Augmented Prototype Explanations for Interactive Medical Image Diagnosis**
R. Khanbayov, E. Serpedin, **H. Kurban**†.
40th Conference on Neural Information Processing Systems (NeurIPS), Sydney, Australia, 2026.

S2 **IRIS: A Real-World Benchmark for Inverse Recovery and Identification of Physical Dynamic Systems from Monocular Video**
R. Khanbayov*, M. R. Barhdadi*, E. Serpedin, **H. Kurban**†.
19th European Conference on Computer Vision (ECCV), Malmö, Sweden, 2026.

S3 **Multilingual Prompt Localization for Agent-as-a-Judge: Language and Backbone Sensitivity in Requirement-Level Evaluation**
A. Mahmood*, S. Abdaljalil*, **H. Kurban**†.
3rd Conference on Language Modeling (COLM), San Francisco, CA, USA, 2026.

AI Conferences, Refereed Tracks and Findings

S4 **Perception, Not Reasoning, Limits Visual Theory of Mind**
M. R. Barhdadi, S. T. Wasim, J. Gall, E. Serpedin, **H. Kurban**†.
40th Conference on Neural Information Processing Systems (NeurIPS, Evaluations and Datasets Track), Sydney, Australia, 2026. INVITED ORAL, BABYVLM WORKSHOP

S5 **How Far Can You Grow? Characterizing the Extrapolation Frontier of Graph Generative Models for Materials Science**
C. Polat, E. Serpedin, M. Kurban†, **H. Kurban**†.
32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD, AI for Science Track), Jeju, South Korea, 2026.

S6 **EMPATHIA: Multi-Faceted Human–AI Collaboration for Refugee Integration**
M. R. Barhdadi, M. Tuncel, E. Serpedin, **H. Kurban**†.
39th Conference on Neural Information Processing Systems (NeurIPS, Creative AI Track), San Diego, CA, USA, 2025.

S7 **SAFE: A Sparse Autoencoder-Based Framework for Robust Query Enrichment and Hallucination Mitigation in LLMs**
S. Abdaljalil, F. Pallucchini, A. Seveso, **H. Kurban**†, F. Mercorio, E. Serpedin.
30th Conference on Empirical Methods in Natural Language Processing (EMNLP, Findings), Suzhou, China, 2025.

Journal Articles

S8 **CliffordIP: Clifford Algebra Equivariant Interatomic Potentials for Heterogeneous Catalysis**
C. Polat, E. Serpedin, M. Kurban†, **H. Kurban**†.
npj Computational Materials, 2026. Accepted, in press.

S9 **Linear-Complexity Unified Defense Against Deception Attacks in Distributed Economic Dispatch Using Cryptography and Machine Learning**
M. M. Islam, A. Takiddin, M. Ismail, **H. Kurban**, E. Serpedin.
IEEE Transactions on Smart Grid, 2026. Early access.

S10 **Multimodal Explainable Artificial Intelligence-Driven Analysis of Quantum Size Effects in Copper Nanoclusters for Hydrogen Storage**
M. Kurban, C. Polat, E. Serpedin, **H. Kurban**.
International Journal of Hydrogen Energy, 201, 152950, 2026.

S11 **Bayesian Probabilistic Knowledge from Diameter Prior for Decision Fusion to Detect Lung Nodule Heterogeneity**
M. R. Islam, M. K. Hasan, **H. Kurban**†, E. Serpedin.
IEEE Transactions on Artificial Intelligence, 7(8), 4518–4532, 2026.

S12 **Explainable Machine Learning Reveals Orbital Design Rules for Reversible H_2 Adsorption on Single-Atom-Doped TiO_2 Nanoparticles**
M. Kurban, C. Polat, E. Serpedin, **H. Kurban**.
Materials & Design, 267, 116401, 2026.

S13 **Geometric-k-means: A Bound-Free Approach to Fast and Eco-Friendly k-means**
P. Sharma, M. S. Malec, **H. Kurban**†, M. O. Kulekci, M. M. Dalkilic.
Machine Learning, 115(2), 30, 2025.

Workshop Papers

S14 **Understanding the Capabilities of Molecular Graph Neural Networks in Materials Science Through Multimodal Learning and Physical Context Encoding.**
C. Polat, **H. Kurban**†, E. Serpedin, M. Kurban.
42nd IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR, MM4Mat Workshop), Nashville, TN, USA, 2025. SPOTLIGHT

S15 **Data Expressiveness and Its Use in Data-Centric AI**
H. Kurban†, P. Sharma, M. M. Dalkilic.
35th Conference on Neural Information Processing Systems (NeurIPS, Data-Centric AI Workshop), virtual, 2021.

S16 **xChemAgents: Agentic AI for Explainable Quantum Chemistry**
C. Polat, M. Tuncel, M. Kurban, E. Serpedin, **H. Kurban**†.
42nd International Conference on Machine Learning (ICML, Multi-Agent Systems in the Era of Foundation Models Workshop), Vancouver, Canada, 2025.

S17 **Theorem-of-Thought: A Multi-Agent Framework for Abductive, Deductive, and Inductive Reasoning in Language Models**
S. Abdaljalil, **H. Kurban**†, K. Qaraqe, E. Serpedin.
63rd Annual Meeting of the Association for Computational Linguistics (ACL, Towards Knowledgeable Foundation Models Workshop), Vienna, Austria, 2025.

PUBLICATIONS

Entries are numbered in reverse chronological order within each list. **Bold** denotes Dr. Kurban, a dagger (†) the corresponding author, an asterisk (*) equal contribution, and underlining a student or postdoctoral researcher supervised or co-supervised by Dr. Kurban.

Peer-Reviewed Journal Articles (58)

2026

[58] **When Do Machine-Learned Exchange–Correlation Improvements Inherit into Density-Functional Tight Binding?**
C. Polat, M. Kurban, E. Serpedin, **H. Kurban**†.
Journal of Chemical Theory and Computation, 2026. Accepted, in press.

[57] **Topical Phase Transitions in Artificial Intelligence Research: Large-Scale Evidence and an Early-Warning Signature for Emerging Topics**
R. Khanbayov, **H. Kurban**†.
Quantitative Science Studies, 2026. Accepted, in press.

[56] **Physics as the Label for Measuring and Correcting Materials Reasoning in Multimodal Models**
H. Kurban†, R. Khanbayov, M. Kurban.
Knowledge-Based Systems, 2026. Accepted, in press.

[55] **Embedding Consistent Recurrent Neural Networks for Fracture Modelling in Large Scale Shell Structures**
M. A. Yatkin, M. Kõrgesaar, **H. Kurban**.
Applied Sciences, 16(21), 10718, 2026.

[54] **HalluVerse3: A Fine-Grained Multilingual Benchmark for Hallucination Detection in LLMs**
S. Abdaljalil, E. Serpedin, **H. Kurban**†.
Language Resources and Evaluation, 2026. Accepted, in press.

[53] **A Multimodal Interpretable Approach to Lung Nodule Diagnosis from CT with Radiology and Anatomy-Aware Text**
M. R. Islam, M. Y. Ansari, **H. Kurban**†, E. Serpedin.
IEEE Transactions on Radiation and Plasma Medical Sciences, 2026. Accepted, in press.

[52] **Physics-Guided Thermodynamic Screening of H_2 Adsorption on Alkaline-Earth-Modified Anatase TiO_2 Nanoparticles with LLM-Assisted Interpretation**
M. Kurban, C. Polat, E. Serpedin, **H. Kurban**.
RSC Advances, 2026. Accepted, in press.

[51] **Local Motif-Guided Discovery of Hydrogen-Stabilizing Environments in Refractory High-Entropy Alloys**
M. Kurban, E. Serpedin, C. Polat, **H. Kurban**.
Computational Materials Science, 275, 115094, 2026.

[50] **SCALAR: Quantifying Structural Hallucination, Consistency, and Reasoning Gaps in Materials Foundation Models**
C. Polat, E. Serpedin, M. Kurban, **H. Kurban**†.
Digital Discovery, 2026. Accepted, in press.

[49] **QuantumCanvas: A Multimodal Benchmark for Learning Two-Body Quantum Interactions**
C. Polat, M. Kurban, E. Serpedin, **H. Kurban**†.
Machine Learning: Science and Technology, 7(5), 050502, 2026.

[48] **CliffordIP: Clifford Algebra Equivariant Interatomic Potentials for Heterogeneous Catalysis**
C. Polat, E. Serpedin, M. Kurban†, **H. Kurban**†.
npj Computational Materials, 2026. Accepted, in press.

[47] **Materials Foundation Models Fail Together: Correlated Errors and Collective Blind Spots**
H. Kurban†, M. Kurban.
Computational Materials Science, 2026. Accepted, in press.

[46] **Hydrogen Storage in TiFe Nanoparticles: Void-Limited Saturation and Metal-Selective Charge Transfer During Sequential Loading**
M. Kurban, E. Serpedin, **H. Kurban**.
International Journal of Hydrogen Energy, 250, 156120, 2026.

[45] **Explainable Machine Learning Reveals Orbital Design Rules for Reversible H_2 Adsorption on Single-Atom-Doped TiO_2 Nanoparticles**
M. Kurban, C. Polat, E. Serpedin, **H. Kurban**.
Materials & Design, 267, 116401, 2026.

[44] **Lightweight Defense Against Data Consistency Attacks in Distributed DC Optimal Power Flow**
M. M. Islam, M. Ismail, **H. Kurban**, X. Huo, E. Serpedin.
IEEE Transactions on Automation Science and Engineering, 23, 11658–11669, 2026.

[43] **Descriptor-Guided Thermodynamic Screening of H_2 Adsorption on Single-Atom-Doped Anatase TiO_2 Nanoparticles with Interpretable Machine Learning**
M. Kurban, C. Polat, E. Serpedin, **H. Kurban**.
Scientific Reports, 16(1), 24203, 2026.

[42] **Linear-Complexity Unified Defense Against Deception Attacks in Distributed Economic Dispatch Using Cryptography and Machine Learning**
M. M. Islam, A. Takiddin, M. Ismail, **H. Kurban**, E. Serpedin.
IEEE Transactions on Smart Grid, 2026. Early access.

[41] **Self-Consistent Recurrent Neural Network for Path Dependent Deformation**
M. A. Yatkin, M. Kõrgesaar, V. M. Asan, J. Romanoff, J. Stuckner, **H. Kurban**.
Scientific Reports, 16(1), 20480, 2026.

[40] **SINdex: Semantic INconsistency Index for Hallucination Detection in LLMs**
S. Abdaljalil, P. Sharma, R. Atat, E. Serpedin, **H. Kurban**†.
IEEE Open Journal of the Computer Society, 7, 1145–1155, 2026.

[39] **Bayesian Probabilistic Knowledge from Diameter Prior for Decision Fusion to Detect Lung Nodule Heterogeneity**
M. R. Islam, M. K. Hasan, **H. Kurban**†, E. Serpedin.
IEEE Transactions on Artificial Intelligence, 7(8), 4518–4532, 2026.

[38] **Multimodal Explainable Artificial Intelligence-Driven Analysis of Quantum Size Effects in Copper Nanoclusters for Hydrogen Storage**
M. Kurban, C. Polat, E. Serpedin, **H. Kurban**.
International Journal of Hydrogen Energy, 201, 152950, 2026.

[37] **Engineered MgO Nanoparticles with Tunable Electronic Signatures for Energy Applications**
M. Kurban, C. Polat, E. Serpedin, **H. Kurban**.
ACS Omega, 11(7), 11772–11789, 2026.

2025

[36] **Curriculum-Enhanced Adaptive Sampling for Physics-Informed Neural Networks: A Robust Framework for Stiff PDEs**
H. Cetinkaya, F. Ay, M. Tuncel, H. Nounou, M. N. Nounou, **H. Kurban**, E. Serpedin.
Mathematics, 13(24), 3996, 2025.

[35] **Accelerating Density of States Prediction in Zn-Doped MgO Nanoparticles via Kernel-Optimized Weighted k-NN**
H. Kurban†, P. Sharma, M. M. Dalkilic, M. Kurban.
Scientific Reports, 15(1), 30524, 2025.

[34] **Deep Temporal and Structural Embeddings for Robust Unsupervised Anomaly Detection in Dynamic Graphs**
S. Abdaljalil, **H. Kurban**†, R. Atat, E. Serpedin, K. Qaraqe.
IEEE Open Journal of the Computer Society, 6, 1100–1109, 2025.

[33] **Telescope Indexing for k-Nearest Neighbor Search Algorithms Over High-Dimensional Data & Large Data Sets**
M. Ramachandra, **H. Kurban**†, M. O. Kulekci, M. M. Dalkilic.
Scientific Reports, 15(1), 24788, 2025.

[32] **Geometric-k-means: A Bound-Free Approach to Fast and Eco-Friendly k-means**
P. Sharma, M. S. Malec, **H. Kurban**†, M. O. Kulekci, M. M. Dalkilic.
Machine Learning, 115(2), 30, 2025.

[31] **An Extended Frequency-Improved Legendre Memory Model for Enhanced Long-Term Electricity Load Forecasting**
M. O. Cakiroglu, I. B. Altun, S. R. Fahim, **H. Kurban**†, M. M. Dalkilic, R. Atat, A. Takiddin, E. Serpedin.
IEEE Open Access Journal of Power and Energy, 12, 691–701, 2025.

[30] **Enabling Ease of Access to Quantum Chemistry with Transformer-Based Text Encoding and Physics-Informed Multilayer Perceptron**
C. Polat, **H. Kurban**†, M. Kurban.
Physica Scripta, 100(8), 086009, 2025.

[29] **A Novel Discrete Time Series Representation with De Bruijn Graphs for Enhanced Forecasting Using TimesNet**
M. O. Cakiroglu, **H. Kurban**†, E. K. Buxton, M. M. Dalkilic.
IEEE Access, 13, 123182–123198, 2025.

[28] **CrysMTM: A Multiphase, Temperature-Resolved, Multimodal Dataset for Crystalline Materials**
C. Polat, E. Serpedin, M. Kurban, **H. Kurban**†.
Machine Learning: Science and Technology, 6(3), 030603, 2025.

[27] **QuantumShellNet: Ground-State Eigenvalue Prediction of Materials Using Electronic Shell Structures and Fermionic Properties via Convolutions**
C. Polat, **H. Kurban**†, M. Kurban.
Computational Materials Science, 246, 113366, 2025.

[26] ***p*-ClustVal: A Novel *p*-adic Approach for Enhanced Clustering and Valuation in High-Dimensional scRNASeq Data**
P. Sharma, S. Mishra, **H. Kurban**†, M. M. Dalkilic.
International Journal of Data Science and Analytics, 20(4), 4051–4066, 2025.

2024

[25] **A Noise-Adaptive Machine Learning Framework for Optimizing User Grouping in Dynamic IM-OFDMA Systems**
F. Ay, S. Althunibat, K. Qaraqe, **H. Kurban**.
IEEE Transactions on Communications, 73(3), 1862–1878, 2024.

[24] **A Reinforcement Learning Approach to Effective Forecasting of Pediatric Hypoglycemia in Diabetes I Patients: An Extended de Bruijn Graph**
M. O. Cakiroglu, **H. Kurban**†, K. Qaraqe, L. Aljihmani, G. Petrovski, M. M. Dalkilic.
Scientific Reports, 14(1), 31251, 2024.

[23] **Multimodal Neural Network-Based Predictive Modeling of Nanoparticle Properties from Pure Compounds**
C. Polat, M. Kurban, **H. Kurban**†.
Machine Learning: Science and Technology, 5(4), 045062, 2024.

[22] **Enhancing the Electronic Properties of TiO_2 Nanoparticles Through Carbon Doping: An Integrated DFTB and Computer Vision Approach**
M. Kurban, C. Polat, E. Serpedin, **H. Kurban**.
Computational Materials Science, 244, 113248, 2024.

[21] **An Extended de Bruijn Graph for Feature Engineering Over Biological Sequential Data**
M. O. Cakiroglu, **H. Kurban**†, P. Sharma, M. O. Kulekci, E. K. Buxton, M. Raeeszadeh-Sarmazdeh, M. M. Dalkilic.
Machine Learning: Science and Technology, 5(3), 035020, 2024.

2023

[20] **State of Charge and Temperature-Dependent Impedance Spectra Regeneration of Lithium-Ion Battery by Duplex Learning Modeling**
S. Temiz, S. Erol, **H. Kurban**†, M. M. Dalkilic.
Journal of Energy Storage, 64, 107085, 2023.

2022

[19] **Rapidly Predicting Kohn–Sham Total Energy Using Data-Centric AI**
H. Kurban†, M. Kurban, M. M. Dalkilic.
Scientific Reports, 12, 1–14, 2022.

[18] **Data on Machine Learning Regenerated Lithium-Ion Battery Impedance**
S. Temiz, **H. Kurban**, S. Erol, M. M. Dalkilic.
Data in Brief, 108698, 2022.

[17] **ccImpute: An Accurate and Scalable Consensus Clustering Based Algorithm to Impute Dropout Events in Single-Cell RNA-Seq Data**
M. S. Malec, **H. Kurban**, M. M. Dalkilic.
BMC Bioinformatics, 23, 1–17, 2022.

[16] **Regeneration of Lithium-Ion Battery Impedance Using a Novel Machine Learning Framework and Minimal Empirical Data**
S. Temiz, **H. Kurban**†, S. Erol, M. M. Dalkilic.
Journal of Energy Storage, 52, 105022, 2022.

[15] **DCEM: An R Package for Clustering Big Data via Data-Centric Modification of Expectation Maximization**
P. Sharma, **H. Kurban**, M. M. Dalkilic.
SoftwareX, 17, 100944, 2022.

2021

[14] **Building Machine Learning Systems for Multi-Atoms Structures: $CH_3NH_3PbI_3$ Perovskite Nanoparticles**
H. Kurban†, M. Kurban.
Computational Materials Science, 195, 110490, 2021.

[13] **Predicting Atom Types of Anatase TiO_2 Nanoparticles with Machine Learning**
H. Kurban†, M. Kurban, P. Sharma, M. M. Dalkilic.
Key Engineering Materials, 880, 89–94, 2021.

[12] **Rare-Class Learning Over Mg-Doped ZnO Nanoparticles**
H. Kurban†, M. Kurban.
Chemical Physics, 546, 111159, 2021.

[11] **A DFT Study on Stability and Electronic Structure of AlN Nanotubes**
I. Muz, **H. Kurban**, M. Kurban.
Materials Today Communications, 26, 102118, 2021.

[10] **Effect of Mg Content on Electronic Structure, Optical and Structural Properties of Amorphous ZnO Nanoparticles: A DFTB Study**
H. Kurban, S. Alaei, M. Kurban.
Journal of Non-Crystalline Solids, 560, 120726, 2021.

[9] **Atom Classification with Machine Learning and Correlations Among Physical Properties of ZnO Nanoparticle**
H. Kurban†.
Chemical Physics, 545, 111143, 2021.

[8] **Measuring the Proximity of Medical Treatment Areas with Text Mining**
H. Kurban†.
European Journal of Science and Technology, 21, 518–526, 2021.

2020

[7] **Practical Data Science: Examining the Correlations Between Structural and Electronic Properties of Different Phases of TiO_2 Nanoparticles**
H. Kurban†.
Journal of Selcuk-Technic, 4(19), 1–9, 2020.

[6] **Tailoring the Structural Properties and Electronic Structure of Anatase, Brookite and Rutile Phase TiO_2 Nanoparticles: DFTB Calculations**
H. Kurban, M. M. Dalkilic, S. Temiz, M. Kurban.
Computational Materials Science, 183, 109843, 2020.

2019

[5] **Study of Structural and Optoelectronic Properties of Hexagonal ZnO Nanoparticles**
H. Kurban, M. Kurban.
Bilecik Seyh Edebali University Journal of Science, 6(2), 124–131, 2019.

[4] **Controlling Structural and Electronic Properties of ZnO NPs**
M. Kurban, **H. Kurban**, M. M. Dalkilic.
Bilge International Journal of Science and Technology Research, 3(0), 35–39, 2019.

[3] **Density-Functional Tight-Binding Approach for the Structural Analysis and Electronic Structure of Copper Hydride Metallic Nanoparticles**
H. Kurban, M. Kurban, M. M. Dalkilic.
Materials Today Communications, 21, 100648, 2019.

2017

[2] **Using Data to Build a Better EM: EM* for Big Data**
H. Kurban†, M. Jenne, M. M. Dalkilic.
International Journal of Data Science and Analytics, 4(2), 83–97, 2017.

2014

[1] **Studying the Milky Way Galaxy Using ParaHeap-k, a Parallel Heap-Based *k*-means**
M. Jenne, O. Boberg, **H. Kurban**, M. M. Dalkilic.
IEEE Computer, 47(9), 26–33, 2014.

Peer-Reviewed Conference Proceedings (53)

2027

[53] **BlinkAI: Context-Aware Sentence Generation from Sparse Blink-Selected Input for Augmentative Communication**
S. Abdaljalil, M. Ramachandra, S. Kakatkar, V. Tataroglu, **H. Kurban**†.
41st AAAI Conference on Artificial Intelligence (AAAI, Demonstration Track), Montréal, Canada, 2027.

2026

[52] **GRAPE: Graph-Augmented Prototype Explanations for Interactive Medical Image Diagnosis**
R. Khanbayov, E. Serpedin, **H. Kurban**†.
40th Conference on Neural Information Processing Systems (NeurIPS), Sydney, Australia, 2026.

[51] **Perception, Not Reasoning, Limits Visual Theory of Mind** INVITED ORAL, BABYVLM WORKSHOP
M. R. Barhdadi, S. T. Wasim, J. Gall, E. Serpedin, **H. Kurban**†.
40th Conference on Neural Information Processing Systems (NeurIPS, Evaluations and Datasets Track), Sydney, Australia, 2026.

[50] **Machine-Learning-Assisted Screening of $MgAlH_3$ Perovskite Nanoparticles for Hydrogen Storage in Future Energy Systems**
M. Kurban, E. Serpedin, **H. Kurban**.
52nd Annual Conference of the IEEE Industrial Electronics Society (IECON), Doha, Qatar, 2026.

[49] **Optimizing EV Charging Infrastructure Utilization via Dynamic Rerouting and Predictive Load**
S. R. Fahim, R. Atat, A. Takiddin, M. Ismail, **H. Kurban**, E. Serpedin.
52nd Annual Conference of the IEEE Industrial Electronics Society (IECON), Doha, Qatar, 2026.

[48] **Edge-Conditioned Contrastive Graph Learning for False Data Injection Detection in Transmission Power Systems**
M. Elnour, R. Atat, **H. Kurban**, A. Takiddin, M. Ismail, K. R. Davis, E. Serpedin.
52nd Annual Conference of the IEEE Industrial Electronics Society (IECON), Doha, Qatar, 2026.

[47] **Rotational Robustness and Explainable Machine Learning for Hydrogen Storage Prediction in $LiNH_6$ Nanoparticles**
M. Kurban, E. Serpedin, **H. Kurban**.
52nd Annual Conference of the IEEE Industrial Electronics Society (IECON), Doha, Qatar, 2026.

[46] **Machine Learning-Based Robust Distributed Economic Dispatch Under Data Integrity Attacks**
M. M. Islam, A. Takiddin, M. Ismail, **H. Kurban**, E. Serpedin.
17th IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm), College Station, TX, USA, 2026.

[45] **IRIS: A Real-World Benchmark for Inverse Recovery and Identification of Physical Dynamic Systems from Monocular Video**
R. Khanbayov*, M. R. Barhdadi*, E. Serpedin, **H. Kurban**†.
19th European Conference on Computer Vision (ECCV), Malmö, Sweden, 2026.

[44] **Multilingual Prompt Localization for Agent-as-a-Judge: Language and Backbone Sensitivity in Requirement-Level Evaluation**
A. Mahmood*, S. Abdaljalil*, **H. Kurban**†.
3rd Conference on Language Modeling (COLM), San Francisco, CA, USA, 2026.

[43] **Perception, Not Reasoning, Limits Visual Theory-of-Mind in Multimodal Foundation Models**
M. R. Barhdadi, E. Serpedin, **H. Kurban**†.
43rd IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR, Multi-Modal Reasoning for Agentic Intelligence Workshop), Denver, CO, USA, 2026.

[42] **How Far Can You Grow? Characterizing the Extrapolation Frontier of Graph Generative Models for Materials Science**
C. Polat, E. Serpedin, M. Kurban†, **H. Kurban**†.
32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD, AI for Science Track), Jeju, South Korea, 2026.

[41] **Evaluating Trustworthy Multilingual Alignment in LLMs via Synthetic Natural Language Inference**
S. Abdaljalil, K. Qaraqe, E. Serpedin, **H. Kurban**†.
35th International Joint Conference on Artificial Intelligence and 29th European Conference on Artificial Intelligence (IJCAI-ECAI, Trust-AI Workshop), Bremen, Germany, 2026.

[40] **Monte Carlo Dropout for Robust and Uncertainty-Aware Lung Nodule Classification**
M. R. Islam, C. Mondal, M. Y. Ansari, **H. Kurban**†, E. Serpedin.
34th European Signal Processing Conference (EUSIPCO), Bruges, Belgium, 2026.

[39] **Fair-FL: Verifiable Inclusive Aggregation for Fair Federated Learning**
M. M. Islam, A. Takiddin, M. Ismail, **H. Kurban**, E. Serpedin.
34th European Signal Processing Conference (EUSIPCO), Bruges, Belgium, 2026.

[38] **Strategic Bidding in Competitive and Adversarial Electricity Markets Using No-Regret Learning Algorithms**
M. M. Islam, A. Takiddin, M. Ismail, **H. Kurban**, E. Serpedin.
23rd IFAC World Congress, Busan, South Korea, 2026.

[37] **Site Selectivity Controls Reversible Hydrogen Adsorption and Release on TiFe Nanoparticles: An Interpretable Machine Learning-Assisted Study**
M. Kurban, E. Serpedin, **H. Kurban**.
6th International Conference on Electrical, Computer and Energy Technologies (ICECET), Rome, Italy, 2026.

[36] **Agentic and Explainable Intelligent Screening of Metal–Organic Frameworks for CO_2 Direct Air Capture**
M. Kurban, E. Serpedin, M. Ergun, **H. Kurban**.
8th Intelligent and Fuzzy Systems Conference (INFUS), Ankara, Türkiye, 2026.

[35] **Thermodynamic and Electronic Mapping for Hydrogen Storage on Zr-Decorated TiO_2 Nanoparticles with an Interpretable Learning Agent and Computational Modeling**
M. Kurban, C. Polat, E. Serpedin, **H. Kurban**.
25th International Conference on Advanced Nanomaterials (ANM), Aveiro, Portugal, 2026.

2025

[34] **EMPATHIA: Multi-Faceted Human–AI Collaboration for Refugee Integration**
M. R. Barhdadi, M. Tuncel, E. Serpedin, **H. Kurban**†.
39th Conference on Neural Information Processing Systems (NeurIPS, Creative AI Track), San Diego, CA, USA, 2025.

[33] **Learning Physics Like Humans: Uncovering Physical Laws from Monocular Videos**
M. R. Barhdadi, H. Alnuweiri, **H. Kurban**†.
20th IEEE/CVF International Conference on Computer Vision (ICCV, Human-inspired Computer Vision Workshop), Honolulu, HI, USA, 2025.

[32] **SAFE: A Sparse Autoencoder-Based Framework for Robust Query Enrichment and Hallucination Mitigation in LLMs**
S. Abdaljalil, F. Pallucchini, A. Seveso, **H. Kurban**†, F. Mercorio, E. Serpedin.
30th Conference on Empirical Methods in Natural Language Processing (EMNLP, Findings), Suzhou, China, 2025.

[31] **Stress-Testing Multimodal Foundation Models for Crystallographic Reasoning**
C. Polat, **H. Kurban**†, E. Serpedin, M. Kurban.
63rd Annual Meeting of the Association for Computational Linguistics (ACL, Towards Knowledgeable Foundation Models Workshop), Vienna, Austria, 2025.

[30] **Theorem-of-Thought: A Multi-Agent Framework for Abductive, Deductive, and Inductive Reasoning in Language Models**
S. Abdaljalil, **H. Kurban**†, K. Qaraqe, E. Serpedin.
63rd Annual Meeting of the Association for Computational Linguistics (ACL, Towards Knowledgeable Foundation Models Workshop), Vienna, Austria, 2025.

[29] **xChemAgents: Agentic AI for Explainable Quantum Chemistry**
C. Polat, M. Tuncel, M. Kurban, E. Serpedin, **H. Kurban**†.
42nd International Conference on Machine Learning (ICML, Multi-Agent Systems in the Era of Foundation Models Workshop), Vancouver, Canada, 2025.

[28] **Beyond Atomic Geometry Representations in Materials Science: A Human-in-the-Loop Multimodal Framework**
C. Polat, **H. Kurban**†, E. Serpedin, M. Kurban.
42nd International Conference on Machine Learning (ICML, DataWorld Workshop), Vancouver, Canada, 2025.

[27] **PhysicsNeRF: Physics-Guided 3D Reconstruction from Sparse Views**
M. R. Barhdadi, **H. Kurban**†, H. Alnuweiri.
42nd International Conference on Machine Learning (ICML, Building Physically Plausible World Models Workshop), Vancouver, Canada, 2025.

[26] **Multivariate de Bruijn Graphs: A Symbolic Graph Framework for Time Series Forecasting**
M. O. Cakiroglu, I. B. Altun, M. M. Dalkilic, E. K. Buxton, **H. Kurban**†.
42nd International Conference on Machine Learning (ICML, Foundation Models for Structured Data Workshop), Vancouver, Canada, 2025.

[25] **Understanding the Capabilities of Molecular Graph Neural Networks in Materials Science Through Multimodal Learning and Physical Context Encoding** SPOTLIGHT
C. Polat, **H. Kurban**†, E. Serpedin, M. Kurban.
42nd IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR, MM4Mat Workshop), Nashville, TN, USA, 2025.

[24] **TDCM25: A Multi-Modal Multi-Task Benchmark for Temperature-Dependent Crystalline Materials**
C. Polat, **H. Kurban**†, E. Serpedin, M. Kurban.
13th International Conference on Learning Representations (ICLR, AI for Accelerated Materials Design Workshop), Singapore, 2025.

[23] **De Bruijn Graph-Enhanced Time Series Models for Electricity Load Forecasting**
M. O. Cakiroglu, I. B. Altun, S. R. Fahim, **H. Kurban**†, M. M. Dalkilic, R. Atat, A. Takiddin, E. Serpedin, K. Qaraqe.
17th International Symposium on Signals, Circuits and Systems (ISSCS), Iaşi, Romania, 2025.

[22] **Data-Efficient Hydrogen Adsorption Prediction in Copper Nanoclusters: A Computer Vision-Based Transfer Learning Approach**
C. Polat, **H. Kurban**†, E. Serpedin, M. Kurban.
19th International Conference on Compatibility, Power Electronics and Power Engineering (CPE-POWERENG), Antalya, Türkiye, 2025.

[21] **Instance-Based Learning-Driven Density of States Analysis in Functionalized Fullerene Derivatives for Optimizing Organic Photovoltaics**
P. Sharma, **H. Kurban**†, M. M. Dalkilic, M. Kurban.
19th International Conference on Compatibility, Power Electronics and Power Engineering (CPE-POWERENG), Antalya, Türkiye, 2025.

[20] **Decentralized N-1 Contingency Analysis for Cascading Failure Prediction in Multi-Region Power Systems Using Consortium Blockchain**
M. M. Islam, M. Ismail, R. Atat, **H. Kurban**, K. R. Davis, E. Serpedin.
5th International Conference on Electrical, Computer and Energy Technologies (ICECET), Paris, France, 2025.

[19] **Exploring Various Sequential Learning Methods for Deformation History Modeling**
M. A. Yatkin, M. Kõrgesaar, J. Romanoff, J. Stuckner, Ü. Işlak, **H. Kurban**.
26th International Conference on Engineering Applications of Neural Networks (EANN), Limassol, Cyprus, 2025.

[18] **Predicting Optical Bandgaps in C_{60} and Functionalized Derivatives from Limited Data for Renewable Energy Applications**
M. Tuncel, **H. Kurban**†, E. Serpedin, M. Kurban.
19th International Conference on Compatibility, Power Electronics and Power Engineering (CPE-POWERENG), Antalya, Türkiye, 2025.

2024

[17] **A Novel Discrete Time Series Representation with De Bruijn Graphs for Enhanced Forecasting Using TimesNet** (Extended Abstract).
M. O. Cakiroglu, **H. Kurban**†, E. K. Buxton, M. M. Dalkilic.
11th IEEE International Conference on Data Science and Advanced Analytics (DSAA), San Diego, CA, USA, 2024.

[16] ***p*-ClustVal: A Novel *p*-adic Approach for Enhanced Clustering and Valuation in High-Dimensional scRNASeq Data** (Extended Abstract).
P. Sharma, S. Mishra, **H. Kurban**†, M. M. Dalkilic.
11th IEEE International Conference on Data Science and Advanced Analytics (DSAA), San Diego, CA, USA, 2024.

[15] **What Data-Centric AI Can Do for *k*-means: A Faster, Robust *k*-means-d**
P. Sharma, **H. Kurban**†, M. M. Dalkilic.
41st International Conference on Machine Learning (ICML, DMLR Workshop), Vienna, Austria, 2024.

2023

[14] **Novel NBA Fantasy League Driven by Engineered Team Chemistry and Scaled Position Statistics**
G. Arkanath, N. Gupta, **H. Kurban**†, P. Sharma, M. K. R., E. K. Buxton, M. M. Dalkilic.
11th IEEE International Conference on Big Data (BigData, Data-Centric AI Workshop), Sorrento, Italy, 2023.

[13] **Are Sports Awards About Sports? Using AI to Find the Answer**
A. Shankar, G. V. Rajasekaran, J. Hendricks, J. A. Schlak, P. Sharma, M. K. R., **H. Kurban**†, M. M. Dalkilic.
European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML PKDD, 10th Workshop on Machine Learning and Data Mining for Sports Analytics), Turin, Italy, 2023.

[12] **AReS: An AutoML Regression Service for Data Analytics and Novel Data-Centric Visualizations**
J. Elms, S. Johnson, M. K. R., K. Sugasi, P. Sharma, **H. Kurban**†, M. M. Dalkilic.
29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD, Undergraduate Consortium), Long Beach, CA, USA, 2023.

[11] **Are They What They Claim: A Comprehensive Study of Ordinary Linear Regression Among the Top Machine Learning Libraries in Python**
S. Johnson, J. Elms, M. K. R., K. Sugasi, P. Sharma, **H. Kurban**†, M. M. Dalkilic.
29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD, Undergraduate Consortium), Long Beach, CA, USA, 2023.

2021

[10] **Data Expressiveness and Its Use in Data-Centric AI**
H. Kurban†, P. Sharma, M. M. Dalkilic.
35th Conference on Neural Information Processing Systems (NeurIPS, Data-Centric AI Workshop), virtual, 2021.

2019

[9] **Size Dependent Electronic Structure and Structural Properties of Cupric Oxide (CuO) Nanoparticles**
H. Kurban, M. Kurban, M. M. Dalkilic.
International Natural Science, Engineering and Material Technologies Conference (NEM), Istanbul, Türkiye, 2019.

2018

[8] **Using Data Analytics to Optimize Public Transportation on a College Campus**
K. Zimmer, **H. Kurban**, M. Jenne, L. Keating, P. Maull, M. M. Dalkilic.
5th IEEE International Conference on Data Science and Advanced Analytics (DSAA), Turin, Italy, 2018.

2017

[7] **A Novel Approach to Optimization of Iterative Machine Learning Algorithms: Over Heap Structure**
H. Kurban†, M. M. Dalkilic.
5th IEEE International Conference on Big Data (BigData), Boston, MA, USA, 2017.

[6] **Improving Expectation Maximization Algorithm Over Stellar Data**
H. Kurban†, C. Kockan, M. Jenne, M. M. Dalkilic.
5th IEEE International Conference on Big Data (BigData, Workshop on Management, Search and Mining of Massive Repositories of Solar and Stellar Astronomy Data), Boston, MA, USA, 2017.

[5] **Case Study: Clustering Big Stellar Data with EM*** BEST POSTER AWARD
H. Kurban†, C. Kockan, M. Jenne, M. M. Dalkilic.
4th IEEE/ACM International Conference on Big Data Computing, Applications and Technologies (BDCAT), Austin, TX, USA, 2017.

2016

[4] **Employing Software Engineering Principles to Enhance Management of Climatological Datasets for Coral Reef Analysis**
M. Jenne, A. Zimmerman, **H. Kurban**, C. Johnson, M. M. Dalkilic.
6th International Workshop on Climate Informatics (CI), Boulder, CO, USA, 2016.

[3] **EM*: An EM Algorithm for Big Data** HONORABLE MENTION PAPER AWARD
H. Kurban†, M. Jenne, M. M. Dalkilic.
3rd IEEE International Conference on Data Science and Advanced Analytics (DSAA), Montréal, Canada, 2016.

2015

[2] **Red-RF: Reduced Random Forest for Big Data Using Priority Voting & Dynamic Data Reduction**
H. Mohsen, **H. Kurban**, K. Zimmer, M. Jenne, M. M. Dalkilic.
IEEE International Congress on Big Data (BigData Congress), New York, NY, USA, 2015.

2014

[1] **A New Set of Random Forests with Varying Dynamic Data Reduction and Voting Techniques**
H. Mohsen, **H. Kurban**, M. Jenne, M. M. Dalkilic.
1st IEEE International Conference on Data Science and Advanced Analytics (DSAA), Shanghai, China, 2014.

Book Reviews (3)

[3] *Data Science For All* by B. Davis and H. Glanz, 2024. ISBN: 9780135311189.

[2] *Mastering Social Media Mining with R* by S. K. Ravindran, 2015. ISBN: 1784396311.

[1] *Learning Data Mining with R* by B. Makhabel, 2015. ISBN: 1783982101.

Manuscripts Under Review (83)

Status appears at the right margin: *submitted*, *minor revision*, or *major revision*.

Journal submissions

[83] **SPINDLE: Real-Time Proprioceptive Correction for Execution-Robust Action-Chunked VLAs**
F. Mohsen, T. Elkaffash, **H. Kurban**, M. Mabrok, N. Meskin, A. Safa. *IEEE Robotics and Automation Letters*. *submitted*

[82] **Where the Real Clips Came From: Corpus Identity in AI-Generated Video Detection**
M. O. Cakiroglu, Z. Lu, M. M. Dalkilic, **H. Kurban**. *Transactions on Machine Learning Research*. *submitted*

[81] **ConfTriage: Calibration-Aware Large Language Model Triage for Pulmonary Nodule Malignancy with Selective Specialist Deferral**
M. R. Islam, S. Abdaljalil, E. Serpedin, **H. Kurban**†. *IEEE Journal of Biomedical and Health Informatics*. *major revision*

[80] **The Shape of AI Safety: A Cartographic Synthesis and Analysis of 12,987 Papers from Premier Venues (2018–2026)**
I. B. Altun, E. K. Buxton, M. M. Dalkilic, **H. Kurban**†. *Artificial Intelligence Review*. *major revision*

[79] **Nanoscale Stoichiometry Drives Enhanced Hydrogen Storage in $LiBeH_3$ Perovskite Hydride Nanoparticles**
M. Kurban, E. Serpedin, C. Polat, **H. Kurban**. *International Journal of Hydrogen Energy*. *submitted*

[78] **NanoCIF: Text-Based Autoregressive Generation of Finite Inorganic Nanoparticle Structures**
M. Kurban, E. Serpedin, **H. Kurban**. *Machine Learning: Science and Technology*. *submitted*

[77] **Reusable Density Control for Inverse Materials Design with Autoregressive Models**
C. Polat, M. Kurban†, E. Serpedin, **H. Kurban**†. *Machine Learning: Science and Technology*. *submitted*

[76] **Geometric Algebra Meets Cartesian Tensors: Higher-Order Equivariance for Interatomic Potentials**
C. Polat, M. Kurban†, E. Serpedin, **H. Kurban**†. *Digital Discovery*. *submitted*

[75] **STEMGym: Benchmarking Sequential Decision-Making Under Dose Budgets in Autonomous Electron Microscopy**
C. Polat, M. Kurban, E. Serpedin, **H. Kurban**. *Machine Learning: Science and Technology*. *submitted*

[74] **Hypothetical Labels in Computational Materials Databases Are Not a Reliable Proxy for Compositional Novelty**
H. Kurban†, M. Kurban†. *Nature Communications*. *submitted*

[73] **Sequential Molecular Hydrogen Loading on the ZIF-8 Metal–Organic Framework: Adsorption Hierarchy, Electronic Structure, and Charge Transfer Mechanism**
M. Kurban, E. Serpedin, **H. Kurban**. *International Journal of Hydrogen Energy*. *major revision*

[72] **Agentic AI in Biomedical and Healthcare: A Modality-Centric Review of Emerging Frameworks, Open Challenges, and Future Directions**
M. R. Islam, M. M. Islam, M. Y. Ansari, **H. Kurban**†, E. Serpedin. *Artificial Intelligence Review*. *major revision*

[71] **The Parity Gap in Crystal Tensor Prediction**
C. Polat, M. Kurban, E. Serpedin, **H. Kurban**†. *Scientific Reports*. *submitted*

[70] **Structural Instability of Cubic XNH6 (X = Na, K) Hydrides for Hydrogen Storage: Symmetry-Breaking Collapse in the Bulk and Framework Decomposition in Nanoparticles**
M. Kurban, E. Serpedin, Doru Ursutiu, **H. Kurban**. *Journal of Energy Storage*. *submitted*

[69] **Grounded Verification of Chemical and Materials Reasoning: Detection Is the Bottleneck**
C. Polat, E. Serpedin, M. Kurban, **H. Kurban**†. *Communications AI & Computing*. *submitted*

[68] **Certified Discovery: Statistically Valid Materials Screening Claims from a Small DFT Verification Budget**
H. Kurban†, M. Kurban. *Digital Discovery*. *submitted*

[67] **The Persuasion–Prediction Gap: Outcome-Grounded Auditing of LLM Judges on Market-Resolved Financial and Crypto Reasoning**
V. Tataroglu, **H. Kurban**†. *Expert Systems with Applications*. *submitted*

[66] **Site-Engineered Reversible H_2 Adsorption on Zr-Doped Anatase TiO_2 Nanoparticles: Active Learning-Assisted Hydrogen Storage Screening**
M. Kurban, C. Polat, E. Serpedin, **H. Kurban**. *International Journal of Hydrogen Energy*. *submitted*

[65] **When Filtering Unsupported Premises Improves Language-Model Reasoning**
S. Abdaljalil, E. Serpedin, **H. Kurban**†. *IEEE Transactions on Artificial Intelligence*. *submitted*

[64] **Stealthy, High-Impact False Data Injection Attacks in Smart Grid Using Physics-Constrained GAN**
M. M. Islam, **H. Kurban**, A. Takiddin, M. Ismail, R. Atat, E. Serpedin. *IEEE Transactions on Smart Grid*. *submitted*

[63] **Byzantine-Resilient Distributed EV Charging Using Threshold Cryptography and Machine Learning**
M. M. Islam, M. Ismail, **H. Kurban**, A. Takiddin, E. Serpedin. *IEEE Transactions on Transportation Electrification*. *major revision*

[62] **Byzantine-Robust, Privacy-Preserving Distributed Unit Commitment Using a Hybrid Approach**
M. M. Islam, M. Ismail, **H. Kurban**, E. Serpedin. *IEEE Transactions on Industrial Cyber-Physical Systems*. *submitted*

[61] **Robust Graph Contrastive Learning for Attack Detection in Power Distribution Systems**
M. Elnour, **H. Kurban**, R. Atat, A. Takiddin, M. Ismail, K. R. Davis, E. Serpedin. *IEEE Open Access Journal of Power and Energy*. *submitted*

[60] **Byzantine-Resilient Distributed DC Optimal Power Flow Using Cryptography and Machine Learning**
M. M. Islam, M. Ismail, **H. Kurban**, A. Takiddin, E. Serpedin. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*. *major revision*

[59] **Benchmarking Deep Time-Series Models for Multi-Day Photovoltaic Generation Forecasting Across Input Settings and Weather Regimes**
M. Elnour, M. AlShaikh Saleh, M. O. Cakiroglu, **H. Kurban**, R. Atat, A. Takiddin, M. Ismail, K. R. Davis, E. Serpedin. *Energy and AI*. *submitted*

[58] **Joint Sensor Deployment and Physics-Informed Graph Transformer Optimization for Smart Grid Cyberattack Detection**
M. Elnour, M. AlShaikh Saleh, R. Atat, X. Huo, A. Takiddin, M. Ismail, **H. Kurban**, K. R. Davis, E. Serpedin. *IEEE Access*. *submitted*

[57] **Benchmarking Artificial Intelligence Models for Dissolved Gas Forecasting in Power Transformers**
I. B. Altun, M. O. Cakiroglu, S. Awadallah, M. M. Dalkilic, **H. Kurban**†. *Engineering Applications of Artificial Intelligence*. *submitted*

Conference submissions

[56] **Phantom Robustness: Gradient Evaluations Give False Assurance for Motion-Vector Detectors of AI-Generated Video**
M. O. Cakiroglu, E. K. Buxton, M. M. Dalkilic, **H. Kurban**. *48th IEEE Symposium on Security and Privacy* (S&P), Montréal, Canada, 2027. *submitted*

[55] **Separating Perception from Reasoning in Vision-Language Models: A Model-Free Render Ceiling for Crystal Structures**
C. Polat, M. Kurban, E. Serpedin, **H. Kurban**†. *44th IEEE/CVF Conference on Computer Vision and Pattern Recognition* (CVPR), Seattle, WA, USA, 2027. *submitted*

[54] **VISTA: A Multimodal Egocentric Benchmark for Goal-Oriented Perception and Assistance to Blind and Low-Vision Users**
N. Mosca, M. R. Barhdadi, M. Hamza, J. Yeh, S. B. Shahi, Z. Li, S.-A. Lali, W. Li, **H. Kurban**†, E. Serpedin, G. Luo, Y. Zhang, M. Li, M. Wang. *44th IEEE/CVF Conference on Computer Vision and Pattern Recognition* (CVPR), Seattle, WA, USA, 2027. *submitted*

[53] **Semantic Slot Canonicalization for Test-Time Scaling of Multi-Concept In-Context Image Customization**
J. Jin, **H. Kurban**. *44th IEEE/CVF Conference on Computer Vision and Pattern Recognition* (CVPR), Seattle, WA, USA, 2027. *submitted*

[52] **Hiding Fakes and Framing Real Footage: A Benchmark and a Robust Detector for AI-Generated Political Video**
M. O. Cakiroglu, E. K. Buxton, M. M. Dalkilic, **H. Kurban**. *44th IEEE/CVF Conference on Computer Vision and Pattern Recognition* (CVPR), Seattle, WA, USA, 2027. *submitted*

[51] **Catching and Repairing Parallax Blindness in Feed-Forward 3D Reconstruction**
O. Bicakcioglu, M. M. Dalkilic, **H. Kurban**. *44th IEEE/CVF Conference on Computer Vision and Pattern Recognition* (CVPR), Seattle, WA, USA, 2027. *submitted*

[50] **Deciding Sets and Exact Tests for Comparing Long-Video Frame Selectors**
H. Yazar, M. Tuncel, **H. Kurban**. *44th IEEE/CVF Conference on Computer Vision and Pattern Recognition* (CVPR), Seattle, WA, USA, 2027. *submitted*

[49] **Hand-Body Contact in Tokenized Sign Motion**
A. Mahmood, S. Abdaljalil, **H. Kurban**. *44th IEEE/CVF Conference on Computer Vision and Pattern Recognition* (CVPR), Seattle, WA, USA, 2027. *submitted*

[48] **Paced Test-Time Memory for Long-Sequence Feed-Forward 3D Reconstruction**
R. Khanbayov, M. R. Barhdadi, **H. Kurban**. *44th IEEE/CVF Conference on Computer Vision and Pattern Recognition* (CVPR), Seattle, WA, USA, 2027. *submitted*

[47] **Self-Grounded Zooming for Thinking Vision-Language Models**
R. Khanbayov, **H. Kurban**. *44th IEEE/CVF Conference on Computer Vision and Pattern Recognition* (CVPR), Seattle, WA, USA, 2027. *submitted*

[46] **Counterfactual Environment Branching for Tool-Using Visual Agents**
R. Khanbayov, **H. Kurban**. *44th IEEE/CVF Conference on Computer Vision and Pattern Recognition* (CVPR), Seattle, WA, USA, 2027. *submitted*

[45] **Track the Unseen Through Its Company: Relational Queries for Long-Range 4D Tracking**
R. AlHamidi, R. Khanbayov, E. Serpedin, **H. Kurban**. *44th IEEE/CVF Conference on Computer Vision and Pattern Recognition* (CVPR), Seattle, WA, USA, 2027. *submitted*

[44] **ReadMatch: Complementary Attention and Similarity for Training-Free Video Frame Selection**
M. W. Monteiro, J. Jin, R. Khanbayov, **H. Kurban**. *44th IEEE/CVF Conference on Computer Vision and Pattern Recognition* (CVPR), Seattle, WA, USA, 2027. *submitted*

[43] **The Gradient That Top-k Distillation Never Computes**
R. Khanbayov, **H. Kurban**†. *30th International Conference on Artificial Intelligence and Statistics* (AISTATS), Montréal, Canada, 2027. *submitted*

[42] **Best-of-N Selection with Discrete Judges: Exact Analysis and Rubric Design**
S. Abdaljalil, E. Serpedin, **H. Kurban**†. *30th International Conference on Artificial Intelligence and Statistics* (AISTATS), Montréal, Canada, 2027. *submitted*

[41] **Conformal Coverage Guarantees for Any Video Temporal Grounder**
A. Mohamed, R. Khanbayov, E. Serpedin, **H. Kurban**†. *30th International Conference on Artificial Intelligence and Statistics* (AISTATS), Montréal, Canada, 2027. *submitted*

[40] **Waiting Buys Exactly Two Moments: Endogenous Identification in Prediction-Driven Allocation**
R. AlHamidi, R. Khanbayov, E. Serpedin, **H. Kurban**†. *15th International Conference on Learning Representations* (ICLR), San Francisco, CA, USA, 2027. *submitted*

[39] **Can Class Names Forecast Per-Class Accuracy for Data-Free CLIP Customization?**
P. Zheng, J. Jin, **H. Kurban**†. *15th International Conference on Learning Representations* (ICLR), San Francisco, CA, USA, 2027. *submitted*

[38] **Bounding Retraining Equivalence and the Deletion Floor in Materials Machine Unlearning**
C. Polat, M. Kurban, E. Serpedin, **H. Kurban**†. *15th International Conference on Learning Representations* (ICLR), San Francisco, CA, USA, 2027. *submitted*

[37] **Physical Survival and Provenance Limits on Watermarking AI-Generated Crystals**
C. Polat, M. Kurban, E. Serpedin, **H. Kurban**†. *15th International Conference on Learning Representations* (ICLR), San Francisco, CA, USA, 2027. *submitted*

[36] **Grafts Add Up: Scoring Every Layer Swap in One Forward–Backward Pass**
A. Mahmood, S. Abdaljalil, **H. Kurban**†. *15th International Conference on Learning Representations* (ICLR), San Francisco, CA, USA, 2027. *submitted*

[35] **Testing Whether a Jailbreak Defense Reduces Attainable Risk**
I. B. Altun, M. M. Dalkilic, **H. Kurban**†. *15th International Conference on Learning Representations* (ICLR), San Francisco, CA, USA, 2027. *submitted*

[34] **The Turn Partition Is a Free Parameter in Multi-Turn LLM Evaluation**
I. B. Altun, E. K. Buxton, M. M. Dalkilic, **H. Kurban**†. *15th International Conference on Learning Representations* (ICLR), San Francisco, CA, USA, 2027. *submitted*

[33] **Oracle Gaps in Reliability Coverage: Sampling Noise or Policy Specialization?**
M. O. Cakiroglu, M. M. Dalkilic, **H. Kurban**†. *15th International Conference on Learning Representations* (ICLR), San Francisco, CA, USA, 2027. *submitted*

[32] **Certified Interface Aliases: Exact Collisions in Vision-Language Preprocessing, and When They Exist**
M. O. Cakiroglu, E. K. Buxton, M. M. Dalkilic, **H. Kurban**†. *15th International Conference on Learning Representations* (ICLR), San Francisco, CA, USA, 2027. *submitted*

[31] **Truncation Samplers Are Nested: An Exact Separation Certificate for Comparing Decoding Rules**
R. Khanbayov, **H. Kurban**†. *15th International Conference on Learning Representations* (ICLR), San Francisco, CA, USA, 2027. *submitted*

[30] **Measuring Textual Degradation in Speech LLMs with a Transcript-Matched Control**
S. Abdaljalil, E. Serpedin, **H. Kurban**†. *52nd IEEE International Conference on Acoustics, Speech and Signal Processing* (ICASSP), Toronto, Canada, 2027. *submitted*

[29] **Reading Arity: A Certified Bound for Evidence Aggregation Under Selection Adversaries**
R. Khanbayov, **H. Kurban**†. *26th International Conference on Autonomous Agents and Multiagent Systems* (AAMAS), Hanoi, Vietnam, 2027. *submitted*

[28] **Effort Reallocation in Self-Verifying LLM Agent Teams**
O. Bicakcioglu, M. M. Dalkilic, **H. Kurban**†. *26th International Conference on Autonomous Agents and Multiagent Systems* (AAMAS), Hanoi, Vietnam, 2027. *submitted*

[27] **Language-Conditioned Rank Reversal in Agentic LLM Judges**
A. Mahmood, S. Abdaljalil, **H. Kurban**†. *26th International Conference on Autonomous Agents and Multiagent Systems* (AAMAS), Hanoi, Vietnam, 2027. *submitted*

[26] **Do Gaussian Scenes Contain Enough Structure for Intrinsic Segmentation?**
M. R. Barhdadi, H. Yazar, E. Serpedin, M. Tuncel, **H. Kurban**†. *14th International Conference on 3D Vision* (3DV), Thessaloniki, Greece, 2027. *submitted*

[25] **What Do Similarity-Scored Preference Pairs Select? A Provenance-Anchored Audit of Label-Free DPO**
J. Azem, S. Abdaljalil, **H. Kurban**†. *32nd International Conference on Computational Linguistics* (COLING), Macau, China, 2027. *submitted*

[24] **When Do Errors Compound Across Languages? A Descriptive Study of Multilingual Long-Horizon Tool Use**
A. Mahmood, S. Abdaljalil, **H. Kurban**†. *32nd International Conference on Computational Linguistics* (COLING), Macau, China, 2027. *submitted*

[23] **Separating Knowledge from Application in LLM Judges**
S. Bouhroum, H. Al-Rumaihi, A. Mahmood, **H. Kurban**. *32nd International Conference on Computational Linguistics* (COLING), Macau, China, 2027. *submitted*

[22] **Reward-Model Bias in Multi-Agent LLM Routing**
A. Kannan, E. Serpedin, **H. Kurban**. *32nd International Conference on Computational Linguistics* (COLING), Macau, China, 2027. *submitted*

[21] **Question-Agnostic Crops Trade Evidence for Diversity in Vision-Language Voting**
M. W. Monteiro, R. Khanbayov, J. Jin, **H. Kurban**. *Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics* (NAACL), San Francisco, CA, USA, 2027. *submitted*

[20] **Safe for Whom? A Counterfactual Audit of Demographic Refusal Disparities Under Steering Defenses in Vision-Language Models**
I. B. Altun, E. K. Buxton, M. M. Dalkilic, **H. Kurban**. *10th ACM Conference on Fairness, Accountability, and Transparency* (FAccT), Porto, Portugal, 2027. *submitted*

[19] **Knowing When Not to Answer: Evidence-Derived Abstention Requires Evidence That Matches the Claim**
S. Abdaljalil, E. Serpedin, **H. Kurban**†. *32nd International Conference on Computational Linguistics* (COLING), Macau, China, 2027. *submitted*

[18] **Why Gain Ratios Mislead: Identifiability and Power in Paired LLM Evaluation**
S. Abdaljalil, E. Serpedin, **H. Kurban**†. *32nd International Conference on Computational Linguistics* (COLING), Macau, China, 2027. *submitted*

[17] **Ask the Judge Twice: A Label-Free Certificate for Cross-Lingual Reward Hacking**
A. Mahmood, S. Abdaljalil, **H. Kurban**†. *20th Conference of the European Chapter of the Association for Computational Linguistics* (EACL), Athens, Greece, 2027. *submitted*

[16] **When Does Consensus Mean Correctness? Measuring the Agreement–Accuracy Coupling with Semantics-Preserving Re-Rendering**
R. Khanbayov, **H. Kurban**†. *20th Conference of the European Chapter of the Association for Computational Linguistics* (EACL), Athens, Greece, 2027. *submitted*

[15] **Consistency Has a Computable Blind Spot: A Commutation Theory of Label-Free Reliability for Vision-Language Figure Reading**
R. Khanbayov, **H. Kurban**†. *20th Conference of the European Chapter of the Association for Computational Linguistics* (EACL), Athens, Greece, 2027. *submitted*

[14] **Do Visual Agents Use Retrieved Evidence? A Matched Counterfactual Probe for Video Reasoning**
R. AlHamidi, R. Khanbayov, E. Serpedin, **H. Kurban**†. *Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics* (NAACL), San Francisco, CA, USA, 2027. *submitted*

[13] **MERIT: Multi-Perspective Evidence-Gated Review for ICU Triage**
I. Haseeb, A. Kannan, S. Abdaljalil, E. Serpedin, **H. Kurban**†. *41st AAAI Conference on Artificial Intelligence* (AAAI), Montréal, Canada, 2027. *submitted*

[12] **A Low Grounding Score Is Not an Ungrounded Judge: Identifying the Perceptibility Confound in Multimodal Oversight**
R. Khanbayov, **H. Kurban**†. *41st AAAI Conference on Artificial Intelligence* (AAAI), Montréal, Canada, 2027. *submitted*

[11] **One Perturbation Is Not Enough: Identifiability and Blind Baselines for Behavioral AI Evaluation**
R. Khanbayov, M. Sohail, A. A. Abdala, **H. Kurban**†. *41st AAAI Conference on Artificial Intelligence* (AAAI), Montréal, Canada, 2027. *submitted*

[10] **Thinking Longer, or Told to Think Harder? Instruction Confounding in Test-Time-Compute Evaluation**
I. B. Altun, M. M. Dalkilic, **H. Kurban**†. *41st AAAI Conference on Artificial Intelligence* (AAAI), Montréal, Canada, 2027. *submitted*

[9] **When Does Retrieval Help Time-Series Forecasting**
M. O. Cakiroglu, E. K. Buxton, M. M. Dalkilic, **H. Kurban**†. *41st AAAI Conference on Artificial Intelligence* (AAAI), Montréal, Canada, 2027. *submitted*

[8] **Detect Early, Escalate Rarely: Anytime Detection of AI-Generated Video from the Compressed Bitstream**
M. O. Cakiroglu, M. M. Dalkilic, **H. Kurban**†. *41st AAAI Conference on Artificial Intelligence* (AAAI), Montréal, Canada, 2027. *submitted*

[7] **The Spectrum Is Not Enough: When Context Helps Time-Series Forecasting**
M. O. Cakiroglu, M. M. Dalkilic, **H. Kurban**†. *41st AAAI Conference on Artificial Intelligence* (AAAI), Montréal, Canada, 2027. *submitted*

[6] **4D Synchronized Fields: Motion-Language Gaussian Splatting**
M. R. Barhdadi, S. Abdaljalil, R. Khanbayov, E. Serpedin, **H. Kurban**†. *41st AAAI Conference on Artificial Intelligence* (AAAI), Montréal, Canada, 2027. *submitted*

[5] **PhysWeep: Does a Video Generator Realize the Physics You Ask For?**
R. Khanbayov, **H. Kurban**†. *27th IEEE/CVF Winter Conference on Applications of Computer Vision* (WACV), Buena Vista, FL, USA, 2027. *submitted*

[4] **Accuracy and Cost Claims Do Not Survive Re-Execution in Agentic VideoQA**
R. AlHamidi, A. Mohamed, R. Khanbayov, M. R. Barhdadi, E. Serpedin, **H. Kurban**†. *27th IEEE/CVF Winter Conference on Applications of Computer Vision* (WACV), Buena Vista, FL, USA, 2027. *submitted*

[3] **LGQ: Learnable Geometric Quantization for Image Tokenization**
I. B. Altun, M. O. Cakiroglu, E. K. Buxton, M. M. Dalkilic, **H. Kurban**†. *27th IEEE/CVF Winter Conference on Applications of Computer Vision* (WACV), Buena Vista, FL, USA, 2027. *submitted*

[2] **Local Chemical Environment and Hydrogen Affinity in TiVZrNbHf High-Entropy Alloy Nanoparticles: DFTB Screening and Spin-PBE Validation**
M. Kurban, E. Serpedin, **H. Kurban**. *6th ISC*, Bursa, Türkiye, 2026. *submitted*

[1] **When Chain-of-Thought Corruption Metrics Fail: Ranking Uncertainty, Score Inflation, and Auditable Comparisons**
M. R. D. Meeran, S. Abdaljalil, **H. Kurban**†. *41st AAAI Conference on Artificial Intelligence* (AAAI, SAFE Workshop), Montréal, Canada, 2027. *submitted*

Research Funding and Grants

Summary: 6 external competitive awards as Lead PI or PI since 2025, the five with no other lead investigator totaling approximately $457,000; 2 national research programs as PI, program totals of $4.43M and $3.20M not attributed individually; 2 student research awards as research mentor, $72,135 held by the students; 11 institutional grants listed below: 9 awards and 2 allocations (startup and faculty resource funds).

External Competitive Research Grants (Lead PI / PI)

2026 **QRDI: Future Communications Call (FCC01).** AI-Driven Real-Time Threat Assessment Framework for Entanglement-Based Quantum Key Distribution Systems.

Role: Lead Principal Investigator; Award No. FCC01-0210-260011; Budget: $274,000.

2026 **QRDI: Agentic Artificial Intelligence Grant.** A Governance and Audit Toolkit for the Responsible Adoption of Agentic Artificial Intelligence in Qatar's Financial Sector.

Role: Lead Principal Investigator; Award No. AAIC01-0130-260007; Budget: $41,000.

2026 **QRDI: Agentic Artificial Intelligence Grant.** Agentic AI Assurance for Qatar: Global Governance Mapping, GCC Benchmarking, and a Finance-Focused Policy Toolkit.

Role: Principal Investigator; Award No. AAIC01-0208-260013; Budget: $40,978.

2025 **QRDI: Carbon Capture and Emissions Control Grant.** Real-Time Self-Adaptive and Autonomous Calibration of CO_2 Leak Detection Sensors in the Presence of Impurities Using Wavelet Transform and Machine Learning.

Role: Principal Investigator (Lead PI: M. A. Rahman); Award No. CCEC02-0216-250065 (HBKU EX-QNRF-CCEC-48); Term: March 2026 to March 2028; Total program budget: $299,792.

2025 **QRDI: Modernizing Charitable Sector Grant.** Advancing AI-Driven Governance in Qatar's Charitable Sector: A Trustworthy Arabic LLM for Regulatory Compliance.

Role: Lead Principal Investigator; Award No. MCSC02-0221-250019 (HBKU EX-QNRF-MCSC-12); Budget: $40,999; Term: February to August 2026, completed.

2025 **TÜBİTAK 1001 Scientific and Technological Research Projects Support Program.** Improving Dynamic Fit and Comfort of Prosthetic Sockets Using Wearable Fiber Optic Sensors.

Role: Principal Investigator; Award No. 125E223; Budget: $60,000.

National Research Programs (PI)

Multi-institution programs led by a Lead Principal Investigator at another institution. Dr. Kurban holds the official role of Principal Investigator, joined each program by invitation from the Lead PI on the date shown, and the figure given is the total program budget rather than an amount attributable to his own program.

2023 **QRDI: National Priorities Research Program (NPRP).**
ReSolVE: Resilient Solutions for Vulnerabilities and Emergencies: An Effective National Risk Management Plan for Qatar.

Role: Principal Investigator, joined by invitation 2025 (Lead PI: T. Al-Ansari); Award No. NPRP14C-0909-210008 (HBKU EX-QNRF-NPRPC-8); Term: March 2023 to March 2028; Total program budget: $4,434,601.

2021 **QRDI: National Priorities Research Program Cluster (NPRP-C).**
Multi-layer Cybersecurity and Situational Awareness to Enhance Resiliency in Qatar's Power Grid.

Role: Principal Investigator, joined by invitation 2026; Award No. NPRP12C-0814-190012; Total program budget: $3,204,904.

Student Research Awards (Research Mentor)

Competitive awards held by the student, with Dr. Kurban as the named research mentor.

2026 **QRDI: Graduate Student Research Award (GSRA).** Efficient Test-Time Reasoning in Compressed Vision–Language Models.

Role: Primary Research Mentor; Student: Mr. Marcus Wein Monteiro, enrolling January 2027; Award No. GSRA13-L-0423-260214; Budget: $46,093.48.

2025 **QRDI: Undergraduate Research Experience Program (UREP).** Leveraging Data Analytics for Interpreting Transformer Dissolved Gas Analysis.

Role: Research Mentor; Award No. UREP31-043-2-014; Budget: $26,042; Completed 2026.

Institutional Grants and Internal Funding

2026 **Student Research Experience Grant (SREG),** Texas A&M University at Qatar. Physics-Guided 4D Gaussian Splatting for Sparse-View Dynamic Scene Understanding.

Role: Principal Investigator; Budget: $44,000.

2026 **HBKU Horizon Hub Workshop Series (2nd Cycle),** Office of the VP for Research. Closing the Loop: Agents, Infrastructure, and Evidence in AI-Driven Science (January 20, 2027).

Role: Lead Principal Investigator; Co-organizing entities: QCRI, QEERI, CHLS, CSE; Budget: QAR 120,000.

2026 **HBKU Horizon Hub Workshop Series (2nd Cycle),** Office of the VP for Research. Navigating the Future: Integrating AI into the Shariah Governance of Islamic Finance Institutions (January 27, 2027).

Role: Principal Investigator; Lead PI: Prof. Nasim S. Shirazi (CIS); Budget: QAR 120,000.

2025 **HBKU Horizon Hub 1st Cycle Program,** Office of the VP for Research. AI for Science Symposium (January 21–22, 2026).

Role: Co-Lead Principal Investigator (with S. Chawla); Award No. HBKU-OVPR-HHW-01-16; Budget: QAR 150,000.

2025 **HBKU Thematic Research Grant Program (3rd Cycle).** Optimal AI and Quantum-Secured Wireless–Optical Communication Design for Smart Cities in Qatar.

Role: Co-Lead Principal Investigator (Lead PI: K. A. Qaraqe); Award No. HBKU-OVPR-TG-03-41; Term: November 2025 to October 2027; Budget: $102,740.

2025 **HBKU GPU Compute Grant,** Research Computing Core Group. Compressed Motion Statistics for Reliable Evaluation of Generative Videos.

Role: Lead Principal Investigator; Budget: QAR 7,000.

2024 **Transformative Educational Experience (TEE) Grant,** Texas A&M University at Qatar. Transforming Software Engineering Education through Autograder and LLM Integration.

Role: Lead Principal Investigator; Budget: $10,000.

2024 **Multiversity Academic Grant,** Texas A&M University at Qatar. A Novel, Multi-Institutional Machine Learning for Engineers Course.

Role: Lead Principal Investigator; Budget: $10,000.

2023 **Faculty Resource Allocation (Three-Year),** Texas A&M University at Qatar.

Role: Lead Principal Investigator; Budget: $53,750.

2023 **Research Impact Initiative,** Texas A&M University at Qatar. Leveraging AI, Machine Learning, and Data Analytics to Enhance Qatar's Sustainable Energy, Healthcare, and Security.

Role: Lead Principal Investigator; Budget: $177,000.

2023 **Startup Research Funds (Three-Year),** Texas A&M University at Qatar.
Role: Lead Principal Investigator; Budget: $413,000.

Open-Source Software and Research Tools

Selected releases. Complete repositories for all projects are linked from the associated papers and from the Kurban Intelligence Lab GitHub organization at github.com/KurbanIntelligenceLab.

- **DCEM**: R package for clustering big data via data-centric modification of Expectation Maximization. Published in *SoftwareX*, 17, 100944, 2022.
- **ccImpute**: Consensus-clustering-based imputation of dropout events in single-cell RNA-seq data; R implementation. Published in *BMC Bioinformatics*, 2022; updated version in *SoftwareX*, 2026.
- ***p*-ClustVal**: A novel *p*-adic approach for enhanced clustering and valuation in high-dimensional scRNA-seq data. Associated with *International Journal of Data Science and Analytics*, 20(4), 4051–4066, 2025.
- **Geometric-*k*-means**: Bound-free, energy-efficient *k*-means clustering algorithm. Associated with *Machine Learning*, 115(2), 30, 2025.
- **IRIS**: Real-world benchmark for inverse recovery and identification of physical dynamic systems from monocular video. Associated with ECCV 2026.
- **SINdex**: Semantic inconsistency index for hallucination detection in large language models. Associated with *IEEE Open Journal of the Computer Society*, 2026.
- **HalluVerse3**: Fine-grained multilingual benchmark for hallucination detection in large language models. Associated with *Language Resources and Evaluation*, 2026.
- **VidAudit**: Six-control protocol and toolkit for auditing generalization in AI-generated video detection.
- **STEMGym**: Gymnasium environment for benchmarking dose-efficient autonomous scanning transmission electron microscopy.
- **TDCM25 / CrysMTM / QuantumCanvas**: Multimodal benchmarks for temperature-dependent crystalline materials, materials foundation models, and visual learning of atomic interactions. Associated with ICLR 2025 and MLST 2025.
- **Extended de Bruijn Graph (eDBG)**: Symbolic time-series and biological-sequence representation framework. Associated with *Machine Learning: Science and Technology*, 2024, and *IEEE Access*, 13, 123182–123198, 2025.
- **AReS**: AutoML regression service with data-centric visualizations. Associated with ACM KDD Undergraduate Consortium, 2023.

Invited Talks and Presentations

Conference Presentations: AI and Machine Learning Venues

- *Perception, Not Reasoning, Limits Visual Theory of Mind.* 40th Conference on Neural Information Processing Systems (NeurIPS, Evaluations and Datasets Track), Sydney, Australia, 2026.

- *IRIS: A Real-World Benchmark for Inverse Recovery and Identification of Physical Dynamic Systems from Monocular Video.* 19th European Conference on Computer Vision (ECCV), Malmö, Sweden, 2026.
- *Multilingual Prompt Localization for Agent-as-a-Judge: Language and Backbone Sensitivity in Requirement-Level Evaluation.* 3rd Conference on Language Modeling (COLM), San Francisco, CA, USA, 2026.
- *Perception, Not Reasoning, Limits Visual Theory-of-Mind in Multimodal Foundation Models.* 43rd IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR, Multi-Modal Reasoning for Agentic Intelligence Workshop), Denver, CO, USA, 2026.
- *How Far Can You Grow? Characterizing the Extrapolation Frontier of Graph Generative Models for Materials Science.* 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD, AI for Science Track), Jeju, South Korea, 2026.
- *Evaluating Trustworthy Multilingual Alignment in LLMs via Synthetic Natural Language Inference.* 35th International Joint Conference on Artificial Intelligence and 29th European Conference on Artificial Intelligence (IJCAI-ECAI, Trust-AI Workshop), Bremen, Germany, 2026.
- *EMPATHIA: Multi-Faceted Human–AI Collaboration for Refugee Integration.* 39th Conference on Neural Information Processing Systems (NeurIPS, Creative AI Track), San Diego, CA, USA, 2025.
- *SAFE: A Sparse Autoencoder-Based Framework for Robust Query Enrichment and Hallucination Mitigation in LLMs.* 30th Conference on Empirical Methods in Natural Language Processing (EMNLP, Findings), Suzhou, China, 2025.
- *Theorem-of-Thought: A Multi-Agent Framework for Abductive, Deductive, and Inductive Reasoning in Language Models.* 63rd Annual Meeting of the Association for Computational Linguistics (ACL, Towards Knowledgeable Foundation Models Workshop), Vienna, Austria, 2025.
- *Stress-Testing Multimodal Foundation Models for Crystallographic Reasoning.* 63rd Annual Meeting of the Association for Computational Linguistics (ACL, Towards Knowledgeable Foundation Models Workshop), Vienna, Austria, 2025.
- *xChemAgents: Agentic AI for Explainable Quantum Chemistry.* 42nd International Conference on Machine Learning (ICML, Multi-Agent Systems in the Era of Foundation Models Workshop), Vancouver, Canada, 2025.
- *Multivariate de Bruijn Graphs: A Symbolic Graph Framework for Time Series Forecasting.* 42nd International Conference on Machine Learning (ICML, Foundation Models for Structured Data Workshop), Vancouver, Canada, 2025.
- *PhysicsNeRF: Physics-Guided 3D Reconstruction from Sparse Views.* 42nd International Conference on Machine Learning (ICML, Building Physically Plausible World Models Workshop), Vancouver, Canada, 2025.
- *Beyond Atomic Geometry Representations in Materials Science: A Human-in-the-Loop Multimodal Framework.* 42nd International Conference on Machine Learning (ICML, DataWorld Workshop), Vancouver, Canada, 2025.
- *Understanding the Capabilities of Molecular Graph Neural Networks in Materials Science Through Multimodal Learning and Physical Context Encoding.* 42nd IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR, MM4Mat Workshop), Nashville, TN, USA, 2025.SPOTLIGHT
- *TDCM25: A Multi-Modal Multi-Task Benchmark for Temperature-Dependent Crystalline Materials.* 13th International Conference on Learning Representations (ICLR, AI for Accelerated Materials Design Workshop), Singapore, 2025.

- *Learning Physics Like Humans: Uncovering Physical Laws from Monocular Videos.* 20th IEEE/CVF International Conference on Computer Vision (ICCV, Human-inspired Computer Vision Workshop), Honolulu, HI, USA, 2025.
- *Exploring Various Sequential Learning Methods for Deformation History Modeling.* 26th International Conference on Engineering Applications of Neural Networks (EANN), Limassol, Cyprus, 2025.
- *What Data-Centric AI Can Do for k-means: A Faster, Robust k-means-d.* 41st International Conference on Machine Learning (ICML, DMLR Workshop), Vienna, Austria, 2024.
- *p-ClustVal: A Novel p-adic Approach for Enhanced Clustering and Valuation in High-Dimensional scRNASeq Data.* 11th IEEE International Conference on Data Science and Advanced Analytics (DSAA), San Diego, CA, USA, 2024.
- *A Novel Discrete Time Series Representation with De Bruijn Graphs for Enhanced Forecasting Using TimesNet.* (Extended Abstract) 11th IEEE International Conference on Data Science and Advanced Analytics (DSAA), San Diego, CA, USA, 2024.
- *Novel NBA Fantasy League Driven by Engineered Team Chemistry and Scaled Position Statistics.* 11th IEEE International Conference on Big Data (BigData, Data-Centric AI Workshop), Sorrento, Italy, 2023.
- *Are Sports Awards About Sports? Using AI to Find the Answer.* European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML PKDD, 10th Workshop on Machine Learning and Data Mining for Sports Analytics), Turin, Italy, 2023.
- *AReS: An AutoML Regression Service for Data Analytics and Novel Data-Centric Visualizations.* 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD, Undergraduate Consortium), Long Beach, CA, USA, 2023.
- *Are They What They Claim: A Comprehensive Study of Ordinary Linear Regression Among the Top Machine Learning Libraries in Python.* 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD, Undergraduate Consortium), Long Beach, CA, USA, 2023.
- *Data Expressiveness and Its Use in Data-Centric AI.* 35th Conference on Neural Information Processing Systems (NeurIPS, Data-Centric AI Workshop), virtual, 2021.
- *Using Data Analytics to Optimize Public Transportation on a College Campus.* 5th IEEE International Conference on Data Science and Advanced Analytics (DSAA), Turin, Italy, 2018.
- *A Novel Approach to Optimization of Iterative Machine Learning Algorithms: Over Heap Structure.* 5th IEEE International Conference on Big Data (BigData), Boston, MA, USA, 2017.
- *Improving Expectation Maximization Algorithm Over Stellar Data.* 5th IEEE International Conference on Big Data (BigData, Workshop on Management, Search and Mining of Massive Repositories of Solar and Stellar Astronomy Data), Boston, MA, USA, 2017.
- *Case Study: Clustering Big Stellar Data with EM*.* BEST POSTER AWARD. 4th IEEE/ACM International Conference on Big Data Computing, Applications and Technologies (BDCAT), Austin, TX, USA, 2017.
- *EM*: An EM Algorithm for Big Data.* HONORABLE MENTION PAPER AWARD. 3rd IEEE International Conference on Data Science and Advanced Analytics (DSAA), Montréal, Canada, 2016.
- *A New Set of Random Forests with Varying Dynamic Data Reduction and Voting Techniques.* 1st IEEE International Conference on Data Science and Advanced Analytics (DSAA), Shanghai, China, 2014.

Conference Presentations: Domain and Applied Venues

- *Machine-Learning-Assisted Screening of $MgAlH_3$ Perovskite Nanoparticles for Hydrogen Storage in Future Energy Systems.* 52nd Annual Conference of the IEEE Industrial Electronics Society (IECON), Doha, Qatar, 2026.
- *Optimizing EV Charging Infrastructure Utilization via Dynamic Rerouting and Predictive Load.* 52nd Annual Conference of the IEEE Industrial Electronics Society (IECON), Doha, Qatar, 2026.
- *Edge-Conditioned Contrastive Graph Learning for False Data Injection Detection in Transmission Power Systems.* 52nd Annual Conference of the IEEE Industrial Electronics Society (IECON), Doha, Qatar, 2026.
- *Rotational Robustness and Explainable Machine Learning for Hydrogen Storage Prediction in $LiNH_6$ Nanoparticles.* 52nd Annual Conference of the IEEE Industrial Electronics Society (IECON), Doha, Qatar, 2026.
- *Machine Learning-Based Robust Distributed Economic Dispatch Under Data Integrity Attacks.* 17th IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm), College Station, TX, USA, 2026.
- *Monte Carlo Dropout for Robust and Uncertainty-Aware Lung Nodule Classification.* 34th European Signal Processing Conference (EUSIPCO), Bruges, Belgium, 2026.
- *Fair-FL: Verifiable Inclusive Aggregation for Fair Federated Learning.* 34th European Signal Processing Conference (EUSIPCO), Bruges, Belgium, 2026.
- *Strategic Bidding in Competitive and Adversarial Electricity Markets Using No-Regret Learning Algorithms.* 23rd IFAC World Congress, Busan, South Korea, 2026.
- *Site Selectivity Controls Reversible Hydrogen Adsorption and Release on TiFe Nanoparticles: An Interpretable Machine Learning-Assisted Study.* 6th International Conference on Electrical, Computer and Energy Technologies (ICECET), Rome, Italy, 2026.
- *Agentic and Explainable Intelligent Screening of Metal–Organic Frameworks for CO_2 Direct Air Capture.* 8th Intelligent and Fuzzy Systems Conference (INFUS), Ankara, Türkiye, 2026.
- *Thermodynamic and Electronic Mapping for Hydrogen Storage on Zr-Decorated TiO_2 Nanoparticles with an Interpretable Learning Agent and Computational Modeling.* 25th International Conference on Advanced Nanomaterials (ANM), Aveiro, Portugal, 2026.
- *De Bruijn Graph-Enhanced Time Series Models for Electricity Load Forecasting.* 17th International Symposium on Signals, Circuits and Systems (ISSCS), Iaşi, Romania, 2025.
- *Data-Efficient Hydrogen Adsorption Prediction in Copper Nanoclusters: A Computer Vision-Based Transfer Learning Approach.* 19th International Conference on Compatibility, Power Electronics and Power Engineering (CPE-POWERENG), Antalya, Türkiye, 2025.
- *Instance-Based Learning-Driven Density of States Analysis in Functionalized Fullerene Derivatives for Optimizing Organic Photovoltaics.* 19th International Conference on Compatibility, Power Electronics and Power Engineering (CPE-POWERENG), Antalya, Türkiye, 2025.
- *Decentralized N-1 Contingency Analysis for Cascading Failure Prediction in Multi-Region Power Systems Using Consortium Blockchain.* 5th International Conference on Electrical, Computer and Energy Technologies (ICECET), Paris, France, 2025.
- *Predicting Optical Bandgaps in C_{60} and Functionalized Derivatives from Limited Data for Renewable Energy Applications.* 19th International Conference on Compatibility, Power Electronics and Power Engineering (CPE-POWERENG), Antalya, Türkiye, 2025.

- *Size Dependent Electronic Structure and Structural Properties of Cupric Oxide (CuO) Nanoparticles.* International Natural Science, Engineering and Material Technologies Conference (NEM), Istanbul, Türkiye, 2019.
- *Employing Software Engineering Principles to Enhance Management of Climatological Datasets for Coral Reef Analysis.* 6th International Workshop on Climate Informatics (CI), Boulder, CO, USA, 2016.
- *Red-RF: Reduced Random Forest for Big Data Using Priority Voting & Dynamic Data Reduction.* IEEE International Congress on Big Data (BigData Congress), New York, NY, USA, 2015.

Invited Academic Colloquia and Lectures

- *Data-Centric Machine Learning.* Invited colloquia at San Jose State University, University of Houston, University of Montana, American University of the Middle East, United Arab Emirates University, University of Illinois, Zayed University, University of Missouri, and University of South Carolina, 2022 to 2023.
- *Clustering in Data Science.* Department of Statistics and Data Science, Northwestern University, 2022.
- *Comparison of Machine Learning Algorithms for CuO Nanoparticles.* 4th International Conference on Physical Chemistry and Functional Materials (PCFM21), Elazığ, Türkiye, 2021.
- *Practical Data Science: Examining the Correlations Between Structural and Electronic Properties of Different Phases of TiO_2 Nanoparticles.* International Conference on Advanced Technologies (ICAT), Istanbul, Türkiye, 2020.
- *Predicting Atom Types of Anatase TiO_2 Nanoparticles with Machine Learning.* International Conference on Engineering and Innovative Materials (ICEIM), Singapore, 2020.
- *Introduction to Data Science.* Computer Engineering Department, Izmir Institute of Technology, 2020.
- *Clustering Techniques in Engineering.* Computer Engineering Department, Yildirim Beyazit University, 2019.
- *Data Science Fundamentals.* Informatics Department, Istanbul Technical University, 2018.
- *EM Algorithms for Clustering.* Computer Science Department, Indiana University, 2016.
- *Principal Component Analysis.* Computer Science Department, Indiana University, 2016.
- *Data Structures and Algorithms.* Computer Science Department, Indiana University, 2014.
- *Ensemble Models in Machine Learning.* Computer Science Department, Indiana University, 2013.

Industry, Medical, and Leadership Forums

- Invited talk, workshop on "The Role of AI in Strengthening the Resilience of Future Power Systems," Doha, Qatar, 2025.
- *AI-Driven Innovation in Medicine: Success, Failure & Challenges.* 3rd Surgical Annual Research Day, Sidra Medicine, Doha, Qatar, 2025.
- *BlinkAI: A Non-Verbal Communication Tool for Locked-In Syndrome Patients.* AI and Medicine Symposium, Sidra Medicine, Doha, Qatar, 2025.
- *BlinkAI: A Non-Verbal Communication Tool for Locked-In Syndrome Patients.* Web Summit, Doha, Qatar, 2025.

- *A Reinforcement Learning Approach to Effective Forecasting of Pediatric Hypoglycemia in Diabetes I Patients: An Extended de Bruijn Graph.* AI and Medicine Symposium, Sidra Medicine, Doha, Qatar, 2025.
- *AI-Powered Innovation: Driving Organization.* Future Leaders Programme, Muscat, Oman, 2024.
- *Data Science and Big Data.* Eli Lilly and Company, 2017.

Teaching Experience

Instructor of Record

- **ENGR 110:** Introduction to Programming (Undergraduate), Fall 2024, Spring 2025, Fall 2025, Fall 2026.
- **ICT 695:** Master's Thesis (Graduate), Fall 2026.
- **ELEN 250:** Machine Learning for Electrical Engineering (Undergraduate), Spring 2026.
- **ICT 706:** Special Topics in Vision-Language Models (Graduate), Spring 2026.
- **ECEN 491:** Independent Study: Research (Undergraduate), Spring 2025, Summer 2025, Fall 2025, Spring 2026.
- **ECEN 485:** Independent Study: Directed Studies (Undergraduate), Spring 2026.
- **ECEN 691:** Research (Graduate).
- **ENGR 102:** Engineering Lab I: Computation (Undergraduate), Spring 2024, Fall 2024, Spring 2025.
- **ECEN 210:** Computer Programming and Algorithms (Undergraduate), Fall 2023.
- **ECEN 248:** Introduction to Digital Systems Design (Undergraduate), Fall 2023.
- **CSCI-B 565:** Data Mining (Graduate), Spring 2023.
- **CSCI-B 365:** Introduction to Computers and Programming (Undergraduate), Spring 2023.
- **CSCI-C 241:** Discrete Structures for Computer Science (Undergraduate), Summer 2022, Fall 2022.
- **CSCI-B 365:** Introduction to Data Analysis and Mining (Undergraduate), Spring 2022.
- **CSCI-B 505:** Applied Algorithms (Graduate), Fall 2021.
- **BMH 101:** Algorithms and Programming I (Undergraduate), Spring 2019, Fall 2019.
- **BMH 413:** Artificial Neural Networks (Undergraduate), Fall 2018, Fall 2019.
- **BMH 103:** Introduction to Computer Engineering (Undergraduate), Fall 2018, Fall 2019.
- **BMH 406:** Data Security (Undergraduate), Spring 2019.
- **BMH 104:** Web and Internet Technologies (Undergraduate), Spring 2019.
- **BMH 205:** Data Structures (Undergraduate), Fall 2018.
- **CSCI-B 351:** Elements of Artificial Intelligence (Undergraduate/Graduate), Spring 2018.
- **CSCI-B 365:** Introduction to Data Analysis and Mining (Undergraduate), Spring 2018.
- **CSCI-P 556:** Applied Machine Learning (Graduate), Fall 2017.
- **INFO-I 590:** Online Applied Data Mining (Graduate), Fall 2017.

Associate Instructor

- **CSCI-B 565:** Data Mining (Graduate), Fall 2012, Fall 2013, Spring 2015, Spring 2016, Fall 2016.
- **CSCI-B 351:** Elements of Artificial Intelligence (Undergraduate/Graduate), Spring 2017.

- **CSCI-B 555:** Machine Learning (Graduate), Spring 2013.
- **CSCI-C 343:** Data Structures (Undergraduate/Graduate), Spring 2014.
- **CSCI-B 365:** Seminar in Computer Science: Data Mining (Undergraduate), Fall 2014, Fall 2015.
- **CSCI-B 609:** Topics in Algorithms and Computing Theory (Graduate), Fall 2014.
- **INFO-I 590:** Real World Data Science (Graduate, Online), Summer 2016.

Research Supervision and Mentoring

Summary: 1 postdoctoral researcher in post, 1 completed; 6 Ph.D. researchers in training, 2 graduated, 2 further doctoral committees; 3 M.Sc. researchers in training, 3 graduated; 4 undergraduate researchers, 2 graduated; 38 capstone students across 10 teams, 23 graduated.

Postdoctoral Researchers

Current

- **Dr. Jian Jin** (postdoctoral researcher, HBKU, from September 2026; recruited from Nanjing University). Research areas: generative AI; diffusion and score-based generative models; representation learning for multimodal and scientific data.

Graduated

- **Dr. Mehmet Tuncel** (postdoctoral researcher, graduated). *Current position:* Istanbul Technical University (İTÜ), Türkiye.
  Research areas: Physics-informed neural networks and curriculum-based adaptive sampling for stiff PDEs; agentic and explainable AI for quantum chemistry; multimodal property prediction for functionalized fullerenes.
  Selected outcomes: NeurIPS 2025 (Creative AI Track), ICML 2025 workshop.

Doctoral Researchers

Current: Hamad Bin Khalifa University

- **Rasul Khanbayov** (Ph.D. student, HBKU, admitted January 2026 from the Technical University of Munich; sole supervisor). Label-free reliability and identifiability theory for multimodal evaluation; counterfactual auditing of vision–language judges and overseers; conformal coverage guarantees for video temporal grounding.
  Selected outcomes: NeurIPS 2026, ECCV 2026.

Current: Texas A&M University

- **Can Polat** (Ph.D. candidate; coursework and qualifying requirements complete, defense expected Spring 2027). Equivariant and symmetry-aware interatomic potentials (Clifford algebra, parity constraints); scientific foundation models and AI for science; grounding, hallucination, and consistency auditing of materials foundation models.
  Selected outcomes: *npj Computational Materials*, KDD 2026 (AI for Science Track), workshop papers at ICML 2025, ACL 2025, ICLR 2025, and CVPR 2025 (Spotlight).

- **Samir Abdaljalil** (Ph.D. candidate; coursework and qualifying requirements complete, defense expected Spring 2027). Hallucination detection and mitigation in large language models (SINdex, HalluVerse3, SAFE); sparse-autoencoder feature analysis for LLM safety; evidence-grounded reasoning and abstention.
  Selected outcomes: COLM 2026, Findings of EMNLP 2025, *IEEE Open Journal of the Computer Society*, ACL 2025 workshop; 8 co-authored papers accepted and 15 under review.

DOCTORAL ADVISORY COMMITTEE MEMBER: TEXAS A&M UNIVERSITY

- **Md. Mainul Islam** (Ph.D. candidate; expected defense Spring 2027). Byzantine-resilient and privacy-preserving distributed optimization for power systems; false data injection detection in smart grids.
- **Mariam Elnour** (Ph.D. candidate; expected defense Spring 2027). Graph learning for attack detection in power distribution systems; physics-informed graph transformers for smart grid security.

On both committees I contribute to problem definition, methodology and research planning, and meet with the students regularly.

CURRENT: INDIANA UNIVERSITY BLOOMINGTON

- **Mert Onur Cakiroglu** (Ph.D. candidate). Symbolic graph representations for temporal data (de Bruijn and multivariate de Bruijn graphs); long-horizon and retrieval-augmented time-series forecasting; audit protocols for forecasting claims against strong baselines.
  Selected outcomes: *Scientific Reports*, *Machine Learning: Science and Technology*, ICML 2025 workshop.
- **Idil Bilge Altun** (Ph.D. student). Large-scale scientometric mapping of the AI safety literature; hybrid symbolic–neural representations for sequential data; benchmarking for industrial condition monitoring (dissolved gas analysis, long-term load forecasting).
  Selected outcomes: *IEEE Open Access Journal of Power and Energy*, ISSCS 2025, ICML 2025 workshop.
- **Omer Bicakcioglu** (Ph.D. student). Multi-agentic architectures for collaborative reasoning, planning, and tool use; coordination and communication protocols for heterogeneous LLM and VLM agents; multimodal perception and grounded reasoning with vision–language models.
  Selected directions: heterogeneous agent orchestration, multimodal tool use, long-horizon planning, cross-agent verification, VLM-based reasoning, agentic benchmarking.

GRADUATED

- **Md Rabiul Islam** (Ph.D., defended September 2026; co-supervisor). Dissertation: *Interpretable Lung Nodule Heterogeneity Detection Using Bayesian Probabilistic Knowledge, Multimodal Learning, and LLMs*. Interpretable multimodal medical AI; pulmonary-nodule diagnosis from CT with radiology- and anatomy-aware text fusion; Bayesian decision fusion under diameter priors.
  Selected outcomes: *IEEE Transactions on Artificial Intelligence*, *IEEE Transactions on Radiation and Plasma Medical Sciences*, EUSIPCO 2026.
- **Dr. Parichit Sharma** (Ph.D., graduated; co-supervisor, Indiana University Bloomington). *Current position:* University of Maryland, Baltimore, MD, USA.
  Dissertation: Data-Centric Principles for Machine Learning: From Efficient Clustering to Single-Cell Discovery.
  Research areas: Data-centric machine learning; bound-free and energy-efficient clustering (Geometric-k-means, k-means-d); p-adic valuation for high-dimensional data.
  Selected outcomes: *Machine Learning*, *Scientific Reports*, ICML 2024 workshop; 15 co-authored papers.

Master's Students

CURRENT

- **Alhasan Mahmood** (M.Sc. student, HBKU, completing May 2027). Research areas: multilingual LLM-as-a-judge evaluation; rank reversal and label-free double-centering calibration; cross-lingual reward-hacking certificates.
  Selected outcomes: COLM 2026.
- **Mohamed Rafi Dheen Meeran** (M.Sc. student, HBKU, completing May 2027). Research areas: requirement-level evaluation of LLM agents; judge robustness under prompt and backbone variation; benchmark design for agentic systems.
- **Puzhuo Zheng** (M.Sc. student, HBKU, from Fall 2026, recruited from Northeastern University (China)). Research areas: test-time scaling for vision–language models; decoding-format confounds in consistency-based selection; auditing protocols for multimodal inference-time compute.

GRADUATED

- **Ganesh Arkanath** (M.Sc., graduated). *Current position:* Meta. *Thesis: Advancing NBA Playoffs Prediction and Fantasy Leagues Using Feature Engineering, Deep Learning, and Explainable AI.* Research areas: sports analytics; engineered team-chemistry and scaled-position features; explainable deep learning for outcome prediction.
- **Madhavan Kalkunte Ramachandra** (M.Sc., graduated). *Current position:* ServiceNow. Research areas: high-dimensional similarity search; k-nearest-neighbor acceleration and telescope indexing; AutoML regression services; empirical auditing of production machine learning libraries; scalable ML systems.
- **Jaydeep Chauhan** (M.Sc., graduated). *Current position:* Eli Lilly and Company. *Thesis: Out-of-Distribution Generalization, Spurious Correlations, and Supervised Contrastive Groupwise Learning*. Research areas: out-of-distribution generalization; shortcut and spurious-correlation mitigation; groupwise contrastive learning.

Undergraduate Researchers

CURRENT

- **Mohamed Rayan Barhdadi.** *Thesis: The Physical World as the First Teacher: Structured 4D Scene Representations.* Research areas: physics-guided 3D and 4D reconstruction (PhysicsNeRF, Gaussian splatting); visual theory of mind in multimodal models; inverse identification of physical dynamics from monocular video; human–AI collaboration systems.
  Selected outcomes: ECCV 2026, NeurIPS 2026 (Evaluations and Datasets Track; invited oral, BabyVLM Workshop), NeurIPS 2025 (Creative AI Track), ICML 2025, ICCV 2025, and CVPR 2026 workshops.
- **Rama AlHamidi.** Research areas: agentic video understanding and question answering; counterfactual sensitivity and repairability of replay probes in video agents; cost-aware audit protocols for dynamic tool synthesis.
- **Marcus Wein Monteiro.** Research areas: efficient test-time reasoning for small multimodal models; compute-constrained inference. Holder of a QRDI Graduate Student Research Award; enrolls as an M.Sc. student in January 2027.
- **Aseel Mohamed.** Research areas: conformal prediction and distribution-free coverage guarantees for video temporal grounding; evaluation protocols for agentic video question answering; robotics and embodied perception.

GRADUATED

- **Sam Johnson** (graduated). Projects: AutoML regression benchmarking (AReS); empirical analysis of regression algorithms; data-centric diagnostics.
  Selected outcomes: KDD Undergraduate Consortium 2023 (two papers).
- **Josh Elms** (graduated). Projects: AutoML regression platform; visual analytics for model interpretability; large-scale regression behavior analysis.
  Selected outcomes: KDD Undergraduate Consortium 2023 (two papers).

Capstone and Senior Design Teams

10 teams, 38 students. Six teams of 23 students have graduated; four teams of 15 students are in progress in 2026 to 2027.

- **Aseel Mohamed, Rama AlHamidi, Mustafa Eltayeb.** Project: Atlas: Autonomous Testing Lab for Amine Solvents (in progress, 2026 to 2027).
- **Ahmed Al-Thani, Khalifa Al-Mansoori, Mohammed Baghazal, Ziyad Ghalayini.** Project: A Novel Extended de Bruijn Graph Framework for Hypoglycemia Prediction.
  BEST CAPSTONE PROJECT AWARD
- **Abdulaziz AlBalushi, Mohammed AlRasheed, Nasser AlHababi, Mohammed AlNaimi.** Project: Privacy-Preserving Video-Based Emotion Recognition.
- **Fadwa Abbas, Noor Alabudi, Roudha Al-Khaldi, Sandra Mansour.** Project: A Robotics-Assisted Validation Framework for Wi-Fi-Based Indoor Localization.
  BEST CAPSTONE PROJECT AWARD
- **Iqra Yakub, Bashar Farooq, Ali Al-Majid, Abdellatif Hussine.** Project: A Privacy-Preserving Framework for Elderly Activity Monitoring Using Pose Estimation.
  MULTIPLE BEST CAPSTONE AWARDS
- **Noor Ahmed Al-Thani, Muneera Al-Qahtani, Aljazi Naqadan, Al-reem Al-Baker.** Project: An AI-Based Monitoring System for Long-Term Load Forecasting.
- **AlMaha AlEmadi, Aisha Al-Kuwari, Shahd Al-Kubaisi.** Project: Design and Implementation of a Web-Based Clinical Decision Support Service for Multimodal Lung Cancer Detection.

PROFESSIONAL SERVICE AND LEADERSHIP

Leadership Roles

- **Co-Chair (Ph.D.)**, Postgraduate Admission Committee, Division of Computing and Electrical Engineering, Hamad Bin Khalifa University, 2026 to 2027.
- **Lead Organizer and Lead Principal Investigator**: "Closing the Loop: Agents, Infrastructure, and Evidence in AI-Driven Science" (HBKU Horizon Hub Workshop Series, 2nd Cycle), co-organized with QCRI, QEERI and the College of Health and Life Sciences, Doha, Qatar, January 20, 2027.
- **Co-Lead Organizer and Co-Lead Principal Investigator**: "AI for Science Symposium" (HBKU Horizon Hub, 1st Cycle), international meeting, Doha, Qatar, January 21–22, 2026.
- **Workshop Organizer**: "SynWeb: 1st Workshop on Trustworthy Synthetic Data for the Web: Generation, Verification, and Governance," The Web Conference (WWW), Dublin, Ireland, May 10–14, 2027; co-organized with S. Abdaljalil, B. Raonic, and E. Serpedin.

- **Shared Task Organizer**: "Ayn 2026: Cultural Grounding in Arabic Multimodal Generation and Understanding," ArabicNLP 2026 Shared Tasks Program, co-located with EMNLP 2026, Budapest, Hungary, October 24–29, 2026.
- **Special Session Co-Organizer**: 19th International Conference on Compatibility, Power Electronics and Power Engineering (CPE-POWERENG), Special Session on AI-Driven Innovations in Renewable Energy, Antalya, Türkiye, 2025.
- **Special Session Co-Organizer**: 11th IEEE International Conference on Data Science and Advanced Analytics (DSAA), Special Session on Advancing Materials Science through Data Science, San Diego, CA, USA, 2024.

Editorial and Advisory Roles

2023 to present	Editorial Board Member, *Scientific Reports* (Nature Portfolio, Springer Nature)
2023 to present	Technical Editor, Manning Publications
2023 to 2024	Data Science Advisory Board Member, Pearson Education

Program Committees and Session Chairing

- **Program Committee Member**: 41st AAAI Conference on Artificial Intelligence (AAAI), Montréal, Canada, 2027.
- **Technical Program Committee Member**: IEEE International Conference on Blockchain (IEEE Blockchain), 2026.
- **Technical Program Committee Member**: IEEE Cybermatics Congress, Montbéliard, France, 2026.
- **Program Committee Member**: The Web Conference (WWW, Industry Track), Dubai, UAE, 2026.
- **Program Committee Member**: Basarim High-Performance Computing Conference (2020, 2022, 2024).
- **Session Chair**: "Cyber-Physical Systems and Internet of Things in Industry," 52nd Annual Conference of the IEEE Industrial Electronics Society (IECON), Doha, Qatar, 2026.
- **Session Chair**: 4th Digital Flow Assurance Symposium (HBKU, SLB, Wood, TAMUQ), Doha, Qatar, 2025.
- **Session Chair**: 33rd International Joint Conference on Artificial Intelligence (IJCAI), Jeju, South Korea, 2024.
- **Session Chair**: 11th IEEE International Conference on Data Science and Advanced Analytics (DSAA), San Diego, CA, USA, 2024.
- **Session Chair**: 11th IEEE International Conference on Big Data (BigData), Sorrento, Italy, 2023.

Innovation and Technology Transfer

- **MIJHAR Program, Cycle 2** (2025 to 2026), HBKU Innovation Center, Office of Innovation and Industrial Relations. Completed. An eight-month selective program for researchers developing commercialization plans for their technologies, with an assigned business mentor, industry engagement, and a sales pilot.

Peer Review

CONFERENCES

NeurIPS, ICML, ICLR, AAAI (all tracks), ACM SIGKDD (all tracks), AISTATS, AAMAS, ACL Rolling Review (ARR), EACL, COLM, BMVC, IECON, Basarim High-Performance Computing Conference.

Named a *Silver Reviewer* at ICML 2026, a top-reviewer designation based on area-chair ratings of review quality.

JOURNALS

IEEE Transactions on Artificial Intelligence, *IEEE Transactions on Neural Networks and Learning Systems*, *IEEE Transactions on Circuits and Systems I: Regular Papers*, *Machine Learning: Science and Technology*, *Frontiers in Artificial Intelligence*, *Big Data Mining and Analytics*, *Journal of Chemical Information and Modeling*, *Information Sciences*, *Expert Systems with Applications*, *Engineering Applications of Artificial Intelligence*, *Applied Artificial Intelligence*, *Journal of Medical Internet Research*, *Computers in Biology and Medicine*, *Materials & Design*, *Applied Energy*, *Energy Conversion and Management*, *Journal of Energy Storage*, *eTransportation*, *Electric Power Systems Research*, *International Journal of Electrical Power & Energy Systems*, *Digital Communications and Networks*, *Concurrency and Computation: Practice and Experience*, *Systems Science & Control Engineering*, *Results in Engineering*, *Behaviour & Information Technology*, *International Journal of Human-Computer Interaction*, *Frontiers in Psychiatry*, *IEEE Access*, *Entertainment Computing*, *IEEE Transactions on Consumer Electronics*, *Machine Intelligence Research*, *CAAI Transactions on Intelligence Technology*, *Frontiers of Computer Science*, *Optical and Quantum Electronics*, *Frontiers in Big Data*, *SN Computer Science*, *Sustainable Energy, Grids and Networks*, *ACM Transactions on Intelligent Systems and Technology*, *Pediatric Diabetes*, *Physica B: Condensed Matter*, *Discover Nano*, *Energy and AI*, *IEEE Transactions on Automation Science and Engineering*, *ACM AI Letters*, *ACM Transactions on Multimedia Computing, Communications and Applications*, *Food Bioscience*, *Journal of Imaging Informatics in Medicine*, *International Journal of Hydrogen Energy*, *IEEE Transactions on Industrial Informatics*, *Results in Chemistry*, *IEEE Transactions on Vehicular Technology*, *Tsinghua Science and Technology*, *Silicon*, *Computational Materials Science*, *IEEE Transactions on Image Processing*, *Materials Today Communications*.

Committee Service

- **Hamad Bin Khalifa University**, Division of Computing and Electrical Engineering (2026 to 2027): Curriculum and Quality Assurance Committee; Student Recruitment, Development and Activities Committee.
- **Hamad Bin Khalifa University**: AI for Science Taskforce, member.
- **Organizing Committee Member**: "Navigating the Future: Integrating AI into the Shariah Governance of Islamic Finance Institutions" (HBKU Horizon Hub Workshop Series, 2nd Cycle), Doha, Qatar, January 27, 2027.
- **Hamad Bin Khalifa University** (Fall 2024 to Spring 2026): Industrial & Government Outreach Committee; ABET & Curriculum; Academic Advising; AI for Science Initiative.
- **Texas A&M University at Qatar** (Spring 2024): ABET & Curriculum; Seminars & Invited Speakers; Budget.
- **Texas A&M University at Qatar** (Fall 2023): ABET & Curriculum; Seminars & Invited Speakers; Budget; Teaching Load Model.

ACADEMIC AND PROFESSIONAL APPOINTMENTS

2024 to present	**Assistant Professor**, Division of Computing and Electrical Engineering College of Science and Engineering, Hamad Bin Khalifa University, Doha, Qatar
2023 to present	**Adjunct Associate Professor**, Computer Science and Data Science Indiana University Bloomington, IN, USA

2024 to present **Adjunct Assistant Professor**, Electrical and Computer Engineering
Texas A&M University at Qatar, Doha, Qatar

2023 to 2024 **Assistant Professor**, Electrical and Computer Engineering
Texas A&M University at Qatar, Doha, Qatar

2021 to 2023 **Visiting Associate Professor**, Computer Science Department
Indiana University Bloomington, IN, USA

2018 to 2021 **Assistant Professor**, Computer Engineering Department
Siirt University, Siirt, Türkiye

2017 to 2018 **Visiting Assistant Professor**, Computer Science Department
Indiana University Bloomington, IN, USA

2015 to 2017 **Undergraduate Research Mentor**, Computer Science Department
Indiana University Bloomington, IN, USA

2015 to 2016 **Data Scientist**, Turbo Appeal, Chicago, IL, USA

CONFERRED ACADEMIC TITLES

2024 **Doçent** (Associate Professor), conferred by the Inter-University Council of Türkiye (Üniversitelerarası Kurul, ÜAK)

EDUCATION

2017 **Ph.D. in Computer Science**, Minor in Statistics
Indiana University Bloomington, IN, USA
Dissertation: *A Novel Approach to Optimization of Iterative Machine Learning Algorithms: Over Heap Structure*
Advisor: Prof. Mehmet M. Dalkilic
Committee: Prof. Predrag Radivojac, Prof. Michael W. Trosset, Prof. Yuzhen Ye

2012 **M.Sc. in Computer Science**
Indiana University Bloomington, IN, USA

2010 **Certificate in Business Communication**
University of Connecticut, Storrs, CT, USA
Non-degree professional certificate program

2008 **B.Sc. in Mathematics**
İnönü University, Malatya, Türkiye

HONORS AND AWARDS

2026 SILVER REVIEWER: 43rd International Conference on Machine Learning (ICML), awarded on area-chair ratings of submitted reviews

2026 EDITORIAL CONTRIBUTION AWARD: Springer Nature "Editor of Distinction Awards," for editorial service to *Scientific Reports* (Nature Portfolio)

2025 SPOTLIGHT AWARD: 42nd IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR, MM4Mat Workshop), Nashville, TN, USA

2017 BEST POSTER AWARD: 4th IEEE/ACM International Conference on Big Data Computing, Applications and Technologies (BDCAT), Austin, TX, USA

2016 HONORABLE MENTION PAPER AWARD: 3rd IEEE International Conference on Data Science and Advanced Analytics (DSAA), Montréal, Canada

2016 to 2017 NOMINATION: Researcher of the Year, Indiana University Bloomington

2014 to 2015 NOMINATION: Associate Instructor of the Year, Indiana University Bloomington

2010 to 2012 FELLOWSHIP: Computer Science Graduate Fellowship, Indiana University Bloomington

2009 to 2017 SCHOLARSHIP: Turkish National Ministry of Education Scholarship (full graduate studies funding)

TECHNICAL SKILLS, LANGUAGES, AND MEMBERSHIPS

Machine learning	PyTorch, JAX, TensorFlow, Hugging Face Transformers, scikit-learn, PyTorch Geometric, vLLM.
LLMs and agents	LangChain, LangGraph, LlamaIndex, OpenAI and Anthropic APIs, RAG pipelines, LoRA/QLoRA fine-tuning, LLM evaluation and red-teaming.
Systems and HPC	Python, C/C++, R, MATLAB, SLURM, distributed training (DDP, FSDP), CUDA, Docker, Git, Linux, Weights & Biases, MLflow.
Languages	English (fluent, academic and professional), Turkish (native).
Memberships	IEEE, ACM.